提供価値｜専門性を繋ぎ、顧客価値に変える事業開発

ソフト・ハードの技術理解を基に異なる専門性を顧客価値へ結び付け、サービス設計とアライアンス営業で事業化・拡販に繋ぐ



技術領域の理解		技術を事業へ繋ぐ職能	
技術領域	技術が担う機能	技術とサービスの接続機能	事業・運用への展開例
行動認識・需要予測AI	街の人・モビリティ・設備・環境を観測し、行動や状態をデータとして把握。AIによる認識・予測・判断の基礎を形成。	利用意図・移動需要・行動変化を捉え、予約・配車・施設利用と接続。介入・逸脱・異常を希少事象として記録。	需要創出、先読み配車、利用者フィードバック、サービス利用状況の把握へ展開。需要と運行を同じデータで管理。
安全・運行管理システム	人・車両・設備・インフラの観測情報をつなぎ、状況・リスクを把握して安全行動を支援。	歩行者・車両の行動予測、相互警告、介入判断、事故・インシデント記録を一つの安全改善ループへ接続。	事故予防サービス、安全KPI・SLA、運行停止・再開判断、説明・監査記録へ展開し、継続運行の判断材料を形成。
都市・モビリティデータ基盤	都市・施設・車両・センサー等のデータを統合・共有し、複数サービスが利用できる都市データ基盤を形成。	観測データと管理データを分離し、需要・予約・配車・運行を接続。利用目的・同意・権限・保存・移転履歴を管理。	自治体・企業・交通事業者・施設をつなぐ地域共同運行へ展開。API連携、データガバナンス、役割・責任分担を具体化。
自動運転・継続学習基盤	実走行データから継続的に学習し、認識・判断・知識を拡張。未経験事象を抽出して改善へ反映。	介入・逸脱・再配置・環境変化を抽出し、シナリオ再現・評価・改善候補へ接続。ODDと運用条件を一体で更新。	安全性・運行品質・介入率を継続評価し、適用範囲の段階拡張、運用ルール更新、地域展開の判断へ活用。
開発・品質・運用改善基盤	本番利用環境から得たデータを開発・検証へ戻し、実世界とデジタルを循環させながら継続改善。	走行・環境・車両・V2Xデータを評価へ上げ、検証済みのモデル・地図・運行方針を変更単位として管理。	KPI／SLA評価、変更承認、段階配信、停止・再開、監査・ロールバックを含む開発・品質・運用の閉ループへ展開。
シミュレーション・開発基盤	シミュレーション、開発ツール、実行環境、データを組み合わせ、再現・検証・評価を行う開発基盤。	実データ連携・検証基盤として、End-to-Endデータ、シナリオ再生、SIM／実証評価、セキュリティを共通基盤化。	PoC・実証・本番運用の評価条件を接続し、製品オンボーディング、アセスメント、品質確認、開発資産化へ展開。
ロボット・モビリティ基盤	自動配送ロボット・移動空間ロボットを、街の施設・サービス・運行環境と連携して統合運用。	人・荷物・車両・ロボットをODD・能力・時間条件で割り当て、地域Node・受渡拠点・サービス予約と接続。	共同配送、医療・薬剤配送、移動行政、災害物資輸送を共通運行へ束ね、稼働率・継続性・地域KPIを評価。
車両・エネルギー管理基盤	駐車中のEV群の蓄電池をまとめてエネルギー資源として活用。蓄電エネルギーの利用を最適化し、再エネの出力変動を補完。	移動予定と必要電池残量を捉え、車両群・拠点・電力・水素の需給、PPA・Node運用、低CFP価値を接続。	車両群のエネルギー管理、買物中の補充、サービス拠点運用、需給調整、エネルギー原価・収益・投資判断へ展開。



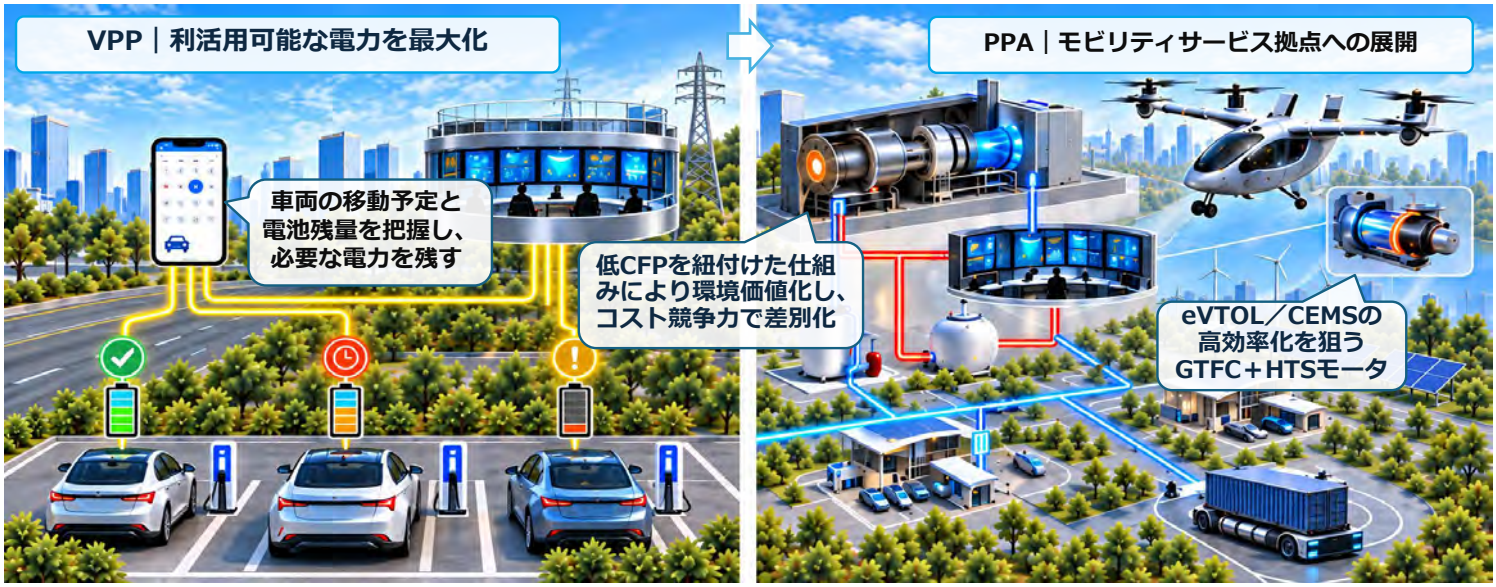
幅広いモビリティ領域の事業化・拡販において、周囲の専門性を繋ぎ、顧客価値と事業成果へ変える役割を担う
推進手法｜顧客にとっての価値から逆算して要素技術を掛け合わせ、単独でも顧客価値が成立する単位に分け、段階的に推進

事業設計例｜プロダクト活用と街事業開発を繋ぐ事業基盤

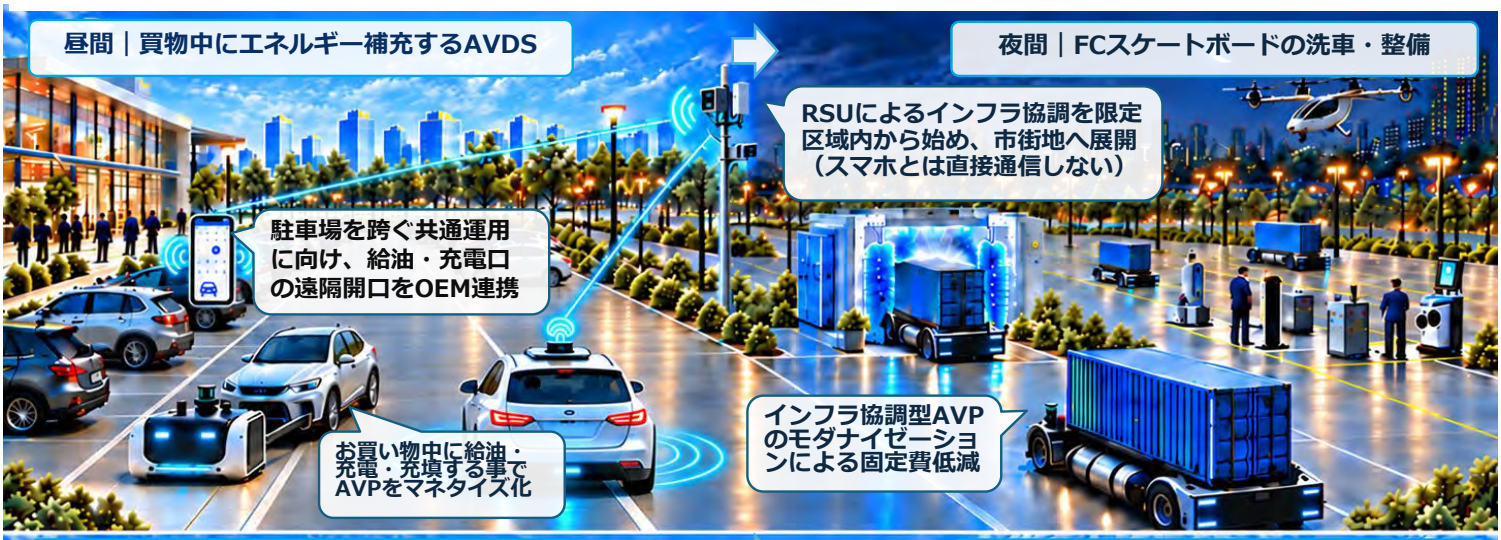
移動・エネルギー利用の個別改善を拠点の共用とサービスの拡張に繋ぐ、モビリティ運行と街事業開発の段階的な事業基盤形成の具体例

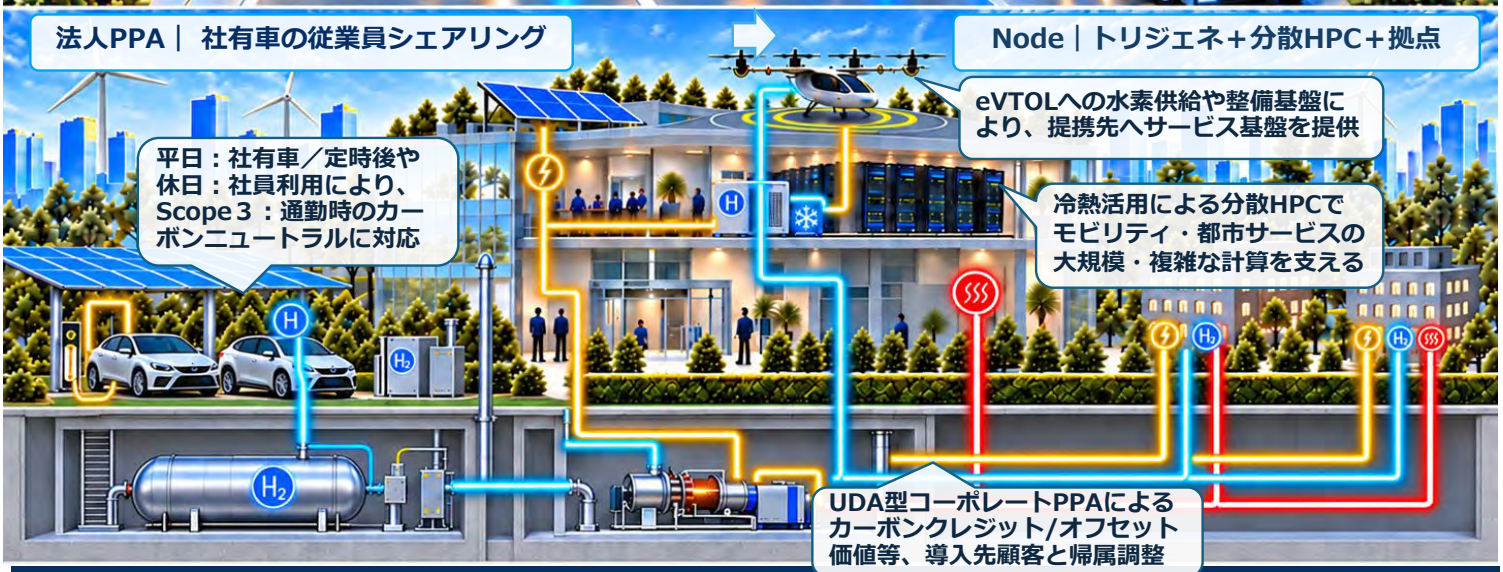
顧客起点｜電力単価の低減とCO2削減要件への対応を導入価値とし、PPAを入口に顧客との合意に基づき顧客便益を維持し街事業へ展開
事業設計｜顧客の導入条件と自社の採算を満たす技術・運用構成を開発側と実現性を検証して具体化し、段階的なサービス拠点化へ繋ぐ

事業基盤｜低CFPを紐付けたVPP基盤 ⇒ PPA展開による自動運転サービス拠点の整備



実装モデル｜大型駐車場／都市ビルに於ける自動運転事業のサービス・事業基盤整備の進め方





!
自動運転の専門軸を起点に、ヒト・モノ・情報・エネルギーへモビリティを拡張し、社会インフラと融合する事業を設計・具体化する
自主開発｜Seneschalにて移動用電力の確保やAVDSサービスの予約を実装／Federationにてサービス運用の縮退制御や事業開発環境を実装

Seneschal｜保険エコシステムを介した相互警告と学習循環

危険兆候を相互警告し、希少事象を品質評価後の学習データ候補として蓄積する

保険会社・自治体・車両を結ぶ相互警告とデータ循環

A 想定主体
- 保険会社
- 自治体
- 自動車メーカー
- Tier1
- 歩行者・車両利用者

B 保険会社への想定価値
- 事故リスク低減
- 損害査定額の検証
- 事故状況分析
- 責任判断の支援
- 契約者視点の強化

C 利用者への想定価値
- 危険接近の補助通知
- 警告・対応履歴検証
- 事故対応支援
- 安全・安心な移動
- 安全サービス活用

D 自動運転開発側の想定価値
- 危険場面候補の蓄積
- 警告ログの蓄積
- 警告後の反応記録
- 警告ロジック改善学習
- データ候補化

人と車の危険兆候を、路側機・クラウド経由で照合する

歩行者の情報(Seneschal)

① 初期|スマホ計測で警告
- 現在位置・進行方向
- 移動履歴・進路変化
- 急停止・急加速
- 危険地点候補の記録
- 受信警告・反応履歴



② 長期|統合データ基盤に記録
- RSU・交通情報と連携
- 信号状態・歩行者検知
- 道路工事・気象情報
- 混雑・天候・視界

③予測精度向上(RSU連携・死角補完)

スマホ×RSU×車両情報を照合し
予測しにくい歩行者行動の兆候を推定
- 急な横断の兆候
- 死角からの飛び出し候補
- 車載センサとの照合
- 予測信頼度の提供



情報として出すもの
- 位置・方向・速度・加速度・履歴
- 過去の行動傾向・危険履歴
- 受信時警告・反応記録・警告回数



車両の情報(AI対応ドラレコ／OBU)

① 初期|AI対応ドラレコで検知・警告
- 接近車両・距離
- 障害物・飛び出し候補
- 車線逸脱・方向指示器
- 急制御・急加速
- 死角・車線変更
- 時刻・位置



② 拡張|ODD条件・車両状態を統合
- 車速・加速度
- 車線位置・ルート
- 制御入力・運転操作
- ADAS・自動運転AIの認識・判断
- OBU通信ログ・時刻同期
- 内部状態

③ 学習用イベント

ドラレコ映像と時刻同期して抽出
- 警告対象・飛び出し
- 反応までの時間
- 通常操作との差分
- 危険領域への進入
- 認識・実走行の差分



1 現在を認識する
人と車の位置・進行方向・速度を同一時刻・地点で照合

2 危険を予測する
人・車・RSUの情報を照合し、予測しにくい行動の兆候と衝突リスクを推定。

3 相互警告する
双方へ補助警告し、危険回避を支援。
※車両制御・安全保証を代替しない

データを価値に変える|保険・安全・自動運転AIへの循環

①イベントデータ候補の収集
BUから危険イベント前後を抽出
- 映像・位置・速度・環境・車両状態
- AI認識・警告・反応を時刻同期
- 同意・匿名化・品質評価後に蓄積

②保険価値を検証
危険場面を分類し、保険サービスへの活用可能性をPoCで評価
- 事故リスク低減効果
- 損害率改善の可能性
- 事故状況の把握支援
- 契約者接点/事故対応

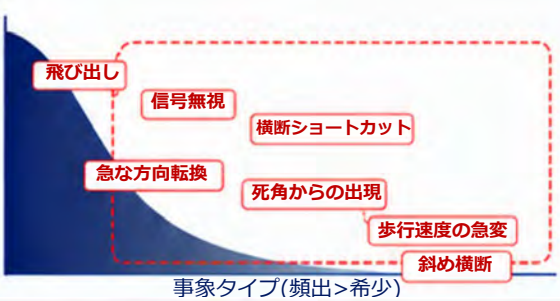
③安全価値を検証
相互警告による危険回避支援と事故リスク低減効果を検証
- 警告ロジック改善
- エリア別リスク評価
- 運用ルール最適化
- 交通安全への活用

④自動運転AI学習へ活用
評価・アノテーション済みイベントを認識・予測モデルの学習候補に
- 認識/予測性能の検証
- 希少事象の補完
- 安全評価への活用

⑤差別化へ変換
歩行者・路側・車載データの補完で単一センサでは得にくい希少事象を収集
- 学習データの多様化
- 警告・予測の改善
- 保険・自治体・OEM連携

実路で得られるイベント
- 夜間・雨・雪・逆光
- 工事・死角・混雑
- 飛び出し・信号無視
- 横断ショートカット
- 進行方向・速度の急変
- 希少な危険・ヒヤリハット

ロングテール事象の整理(歩行者行動に起因する希少事象)



同一時刻・地点の人・車・路側情報を照合
- AI対応ドラレコ/OBU車両
- Seneschal(歩行者)
- RSU(路側機)

複数情報の照合による衝突リスク推定と補助警告

データ源ごとの補完範囲

データ源	主な観測範囲
Seneschal	歩行者の位置・進行方向
AI対応ドラレコ+OBU	車両・運転挙動
RSU	路側・交通渋滞情報
統合照合	単一センサの欠落を相互補完

顧客体験UI
Seneschal

歩行者の路外危険を、車両・RSU・保険会社を結ぶ相互警告とデータ循環で減らし、自動運転学習へ還流する。

https://seneschal.base44.app

後藤悟文

Federation｜街の“今”を捉える分散観測・データ連携

車載・本人同意・道路／施設データを統合し、需要・運行・エネルギーの判断へつなぐ

街を捉える分散観測・データ連携

A. 車載観測・ドラレコ

- V01 道路画像
- V02 GNSS・時間・方位
- V03 速度・加速度
- V04 点群・物体情報
- V05 車番・種別・入口
- V06 行列・渋滞情報
- V07 走行軌跡
- V08 急減速・介入ログ
- V09 センサー品質
- V10 POI変化候補

B. Seneschal（本人同意）

- U01 カレンダー
- U02 現在位置
- U03 行動量
- U04 移動時間帯
- U05 次の予定
- U06 予約状況
- U07 移動履歴
- U08 体調選択
- U09 操作・効率性
- U10 アプリ履歴
- U11 財布動向
- U12 充電選択肢

利用目的・価値説明・アクセス制限を明確化

C. 道路・施設・運行システム

- C01 道路カメラ・信号
- C02 駐車場空き
- C03 充電器空き
- C04 工事・規制
- C05 イベント・施設
- C06 公共交通・遅延
- C07 SOC・蓄電残量
- C08 燃料・電力価格
- C09 営業所・運行DB
- C10 施設予約・利用状況

地域ノード（路側センサー・RSU・エッジ／MEC）による時空間処理



画像処理

- I01 AI／劣化判定
- I02 ﾅﾝﾊﾞｰﾌﾟﾚｰﾄ処理
- I03 点群・座標変換
- I04 物体・道路領域抽出
- I05 特徴量抽出
- I06 品質劣化検知

状態推定

- E01 歩行者数
- E02 走行密度
- E03 混雑度
- E04 駐車場空き
- E05 充電器空き
- E06 工事・規制

時空間処理

- T01 到着時点の状態予測
- T02 駐車場空き予測
- T03 充電器空き予測
- T04 イベント後の人流
- T05 局所降雨確率
- T06 交通・充電不足予測

学習候補抽出

- L01 センサー間整合ログ
- L02 異常事象候補
- L03 高難度事象抽出
- L04 長時間滞留
- L05 ロングテール構造
- L06 モデル更新候補

準リアルタイムStreet ViewとPOI更新情報



街頭の今を画像取得

- R01 駐車場の空き
- R02 充電器の空き
- R03 店舗待ち時間
- R04 日替りメニュー
- R05 駐車場の料金
- R06 GSの価格
- R07 ﾀｸｼｰ待ち人数
- R08 整体の料金

位置・時刻で街の変化を更新

画像×外部情報でPOI化

- P01 店名・施設名
- P02 業種・サービス
- P03 位置・入口
- P04 営業時間・状態
- P05 口コミ・評価
- P06 メニュー料金
- P07 予約・利用条件
- P08 支払条件・ｸｰﾎﾟﾝ

顧客体験への活用

- X01 価格・メニュー・口コミを店舗間で比較
- X02 街の変化から新しい店・サービスを発見
- X03 混雑・行列を避けて待ち時間を短縮
- X04 料金と空き状況からお得な施設を選択

静的地図を進化し、経済活動の街の状態を表す『準リアルタイムStreet View』

学習候補への還流フロー：観測 → 品質・権利確認 → 承認 → 反映

F01 街の“今”を観測・構造化

- F11 道路・入口・施設状態
- F12 走行・軌跡・データ量
- F13 地域ノード時空間処理
- F14 画像・天候・混雑状態
- F15 観測根拠の時刻・場所

F02 学習候補抽出・品質検証

- F21 道路・エリア・用途検証
- F22 車種・POI・入口候補
- F23 変化しやすい名称候補
- F24 画像・権利・個人情報確認
- F25 有用度・信頼度評価

F03 検証・承認後に更新候補へ

- F31 候補選択の充足基準
- F32 品質モデル適合性・有用性
- F33 権利設定・配信モデル検証
- F34 シミュレーション・再生
- F35 承認結果・変更履歴を保存

F04 運行・サービスKPIを評価

- F41 安全性・Safety margin
- F42 効率・POI指標
- F43 経済性・待ち時間
- F44 再走行変化・エネルギーコスト
- F45 変更工数・フィードバック効果

観測 → 候補抽出 → 品質・権利確認 → 再現試験 → 承認 → 段階反映 → KPI評価

Time × Place × Behavior × Observed City State

Seneschal｜予定から移動条件を取得

S01 予定日時	: 2025/5/10（土）12:30
S02 出発地	: 自宅
S03 目的地	: Coffee & Bakery
S04 到着希望時刻	: 12:28
S05 移動目的	: 昼食・買い物
S06 滞在時間	: 60分
S07 同行人数	: 2名
S08 変更可能時間	: ±30分

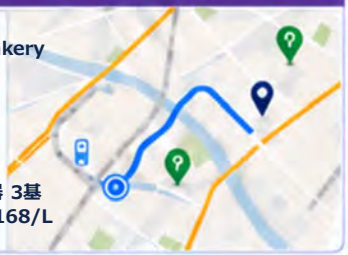
店舗・行列・混雑の観測



- 行列人数 28人
- 待ち時間 約25分
- 混雑度 78%
- 空き駐車場 12台
- 空き充電器 3基

現在位置と経路マップ

- 目的地 Coffee & Bakery
- 到着予想 12:28
- 残り時間 25分
- 駐車場 12台／充電器 3基
- GS（S2）￥168/L



① 時刻別の支援

- S01 現在地・目的地
- S02 次の予定
- S03 所要時間
- S04 駐車・充電空き
- S05 到着希望時刻
- S06 候補経路
- S07 運行経路予測
- S08 料金

予定・移動条件・車の状態を一体化し、経路・出発時刻・手段を支援

② モビリティ需給調整

- N01 タクシー待ち
- N02 人流交通量
- N03 バス運行
- N04 バス待ち
- N05 空車予測
- N06 シェアリング
- N07 イベント需要
- N08 駐車空き
- N09 駐車場EV

需給変動を収集し、運行・配車・供給構造の最適化につなぐ

③ 充電需給調整

- G01 車両SOC
- G02 充電必要性
- G03 充電開始時刻
- G04 到着時刻
- G05 充電器空き
- G06 電力条件
- G07 余剰時間・距離
- G08 充電候補
- G09 充電制約・ピーク

交通需要と電力需給を同一時間軸で予測し、充電配分を最適化

自治体課題に対する活用案

課題	駅前混雑	イベント時の渋滞	観光地渋滞	タクシー不足	高齢者移動	交通空白地	駐車場不足	充電器不足	地域交通赤字	電力ピーク
活用案	人流把握・分散誘導	需要予測・臨時配車	経路分散・交通量調整	需要予測・事前配車	オンデマンド交通支援	路線・乗降拠点設計	空き情報・駐車誘導	空き情報・充電誘導	需要連動運行・効率化	充電時刻の平準化

事業開発環境
Federation

人流・物流・医療・行政を地域Nodeと共通運行基盤で束ね、
地域に必要なサービスを一つの自動運転ループで回す

https://federation.base44.app

Architecture｜実走行データをモデル更新へ返す開発基盤

高価値事象を検証・モデル更新へ返し、安全性とODDを継続的に高める

実走行入力（観測・収集）
RSU観測、OBU走行、車載CAN、センサーデータ、位置・時刻同期

開発資産化（蓄積・整備）
Data Lake、Scenario DB、Edge Case Library、ODD Coverage

モデル改善（検証・改良）
Simulation、SILS／MILS／HILS、Safety Case、OTA

Digital Proving Ground – 実走行を再現し、改善を量産品質へ戻す

① 実走行・観測

- RSU観測：交通流・信号・路面環境
- OBU走行：アプリ・運転操作
- 車載CAN：車両データ・アクチュエータ
- センサーデータ：カメラ・LiDAR・レーダー
- 位置・時刻同期：GNSS／PTP／時刻校正

走行ログ／運転アラート／通信ログ／周辺環境ログ／エッジケース候補

② データ統合・資産化プラットフォーム

- Data Lake：Raw／Processed／Feature Store
- Scenario DB：シナリオ管理・検索
- Edge Case分類：タグ付け・難易度・再現性
- ODD Coverage：空間／道路形状／時間・天候／交通条件／参加者／可視化
- イベント再生（再生再現基盤）：時系列整合／センサ再生／アクチュエータ再現
- モデル評価：故障注入（SOTIF含む）／性能・安全指標／堅牢性（エッジケース）
- パラメータ学習：・シナリオ生成 ・ドメイン適応 ・不確かさ推定
- メタデータ学習：データ系譜／品質指標／権限・監査／ライフサイクル

③ 検証・評価・改善・OTA

- SILS：アルゴ検証
- MILS：ソフト統合検証
- HILS：車両統合検証
- Monte Carlo／Parameter Sweep
- Regression Test：機能リグレッション／性能リグレッション／安全リグレッション／境界条件テスト
- Safety Case：HAZOP／FTA／FMEA／安全目標検証
- Version Management：モデル／データ管理／差分管理・統合コー／トレーサビリティ
- ODD Coverage：カバー率可視化／ギャップ解消／再学習候補
- OTA
- OTA配布・再走行：配信・受信性・整合性検証・再学習

都市デジタルツイン

① 実走行・観測
多様な実走行データを継続的に収集

② データ統合・資産化
統合・整備して再利用できる資産へ

③ シナリオ再現
重要シナリオを再現し検証用データを生成

④ シミュレーション
SILS／MILS／HILSで多面的に検証

⑤ 評価・改善・更新
検証を通じ、モデル／ODD／方針を改善

⑥ OTA配布・再走行
改善内容を配布し、次の実走行で再検証

責任境界／レイヤ責任（共同で改善）

Cloud／City	Policy・KPI・SLA・Governance 合意形成・運用設計・データ配信・説明責任
4D基盤	Data Lake・Scenario DB・Validation データ統合・再現・検証・可視化
MEC／RSU	観測収集・前処理・通信最適化 エッジ性能・時刻同期・品質指標
OBU／Vehicle	記録・共有・暗号・整合・H／W 乗員AI・安全監視・冗長・フェイルセーフ

Real World：実走行での学習と改善

- Seneschal：実運用・調査・インシデント対応・ルール更新
- 道路・信号・交通：道路構造・信号現示・交通流・イベント
- 歩行者・施設・エネルギー：歩行者・自転車・施設・インフラ・充電
- 実走行へ戻す：検証済みPolicy・Map・モデルを安全に配布し、次走行で再学習・更新

Data Up（学習・更新）
Control Down（配布・制御）

計算基盤スタック（概観）

- Sensing：カメラ／LiDAR／ミリ波／GNSS／IMU
- Localization：高精度測位／地図照合
- Perception：物体認識／識別／追跡／予測
- Planning：経路計画／行動生成
- Control：ステア／加減速／制御記録
- MEC／Edge Compute：低遅延／近傍計算
- Fleet Orchestration：配車／運行管理
- Safety／OTA：主張／根拠／証拠／変更記録

計算・通信インフラ領域（適用環境：公道・混在交通・都市）

古典計算（汎用・混合・スケールアウト）
適用領域（演算・ストレージ／通信）
- 高信頼性（冗長アーキテクチャ）
- SBC／x86（TCO最適化）
- 近傍演算力（冷却最適化）
- 高信頼通信（5G／Wi-Fi／衛星）
- データレイク／分析基盤

主な計算資源
- CPU／GPU／NPU／FPGA
- 分散ファイル／コンテナ
- 冗長ストレージ／監視・運用

光コンピューティング
研究・実証／高速化・組合せ問題
適用領域（高速演算・最適化）
- 大規模最適化（空間・制御）
- 組合せ探索（経路・設計）
- 機械学習（行列演算加速）
- 転移学習・高速探索・MPC
- 探索空間の高速評価

主な計算資源
- 光演算加速器／実験チップ
- 光モジュール／ニューラル素子
- 冷却・パッケージング

光電融合／通信・I/O
低遅延・中距離・長距離通信
適用領域（I/O・通信・センサーIF）
- 光インターコネクト／トランシーバ
- センサー・ネットワークI/F
- 大容量通信（データセンター）
- 長距離通信（FSO／Li-Fi）
- エッジ・クラウド連携（DCI）

主な資源
- シリコンフォトニクス／PIC
- 光モジュール／光スイッチ
- 光I/O／CXL over Optics

量子支援
研究・先端／探索支援
量子アニーリング／QML
- 組合せ最適化（ルート・配置）
- 不確かさ推定・確率計算
- 強化学習の高速探索
- 安全性学習の支援

主な資源
- 量子アニーラ／量子シミュレータ
- ハイブリッド（古典＋量子）
- ソフトウェアスタック(QDK/SDK)

高価値データを再利用
実走行の希少イベントを見落とさず、再利用に回す。安全性・コストを最適化。

モデル更新速度向上
再現・検証・更新を高速化し、モデル更新のサイクルを短縮。アジャイル開発を実現。

再現性と説明責任の向上
再現可能な証跡とSafety Caseで、法令や規則に適合する説明責任を確保。

ODD拡張の加速
ODD Coverageを可視化し、不安定領域を集中的に学習。安全領域を継続的に拡張。

量産品質の継続的向上
全社の基準を下げずに、変化の検知で再現性・効率・量産品質を継続的に向上。

街で得た高価値事象を検証・更新へ戻し、次の自治体サービス実装を安全に拡張する。

Seneschal｜24H自動運転で地域生活サービスを再設計

介護・学童・夜間サービスの需要を束ね、人口減少下の生活制約を解く

自治体が放置できない3つの生活制約

少子化
女性就業が標準化しても、送迎・見守りが家庭に残り、就業継続・出生率を阻害

高齢化
1対1訪問と固定デイケアが、介護人材と高齢者の外出機会を奪う

孤独
核家族化や隣近所に誰が住んでいるかも知らず、地域の交流が希薄化する事で犯罪・孤独死等も増加

日中｜移動式介護サービス



- 介護職員が複数人を同時に見守る
- 共通目的地毎に集まり、自由に外出
- 人材不足を緩和し移動の自由を回復

放課後～夜｜移動式学童



- 保育者が運転せず、子どもを見守る
- 学校→習い事→自宅まで通しで預かる
- 就業・第2子・転入定住を支援

夜～深夜3時｜移動式バー



- バーテンダーが接客しながら住宅地へ運行
- 高価な土地代を省き、安価な交流の場へ
- 通勤・飲酒運転を減らし、夜間消費を増幅

08:00–15:00 介護 → **15:00–21:00 学童** → **21:00–03:00 バー** → **03:00–08:00 物流・清掃・整備**

サービスコンテナを取り替えても、車両・運行・予約・Node・データ基盤は共用

三つの専用事業ではない。三つのサービスコンテナで、一つの自動運転基盤の需要の谷を埋める

実行フロー（24時間を支える一つの運営基盤）

1. 需要把握・営業（Seneschal）
2. 予約・認証・決済・マッチング
3. 配車・ルート・遠隔監視（Federation）
4. 交換・清掃・充電（Node）
5. 政策KPIを自治体へ還元

自治体｜施設投資を抑え、複数改革を一つの運行基盤で実行
- 財政健全化・維持費抑制
- 福祉・教育・地域振興を同時推進
- データで施策最適化

住民・事業者｜移動とサービスを一体化し、待ち時間をサービス利用時間へ転換
- 移動時間と待ち時間の短縮
- 固定費・運営の最適化
- 利用機会・地域連携の活性化

学習データ｜時間帯別需要・稼働・運行・介入・エネルギーを基盤取得
- サービス改善を高速化
- 需要予測・車両最適化を高度化
- エネルギーと充電計画を最適化

サービス利用者メリット

- 子育て世帯｜仕事が子育てから遠ざからずに済む
- 高齢者　｜介護を受けても行き先を自分で選べる
- 働く人　｜帰宅方面が同じ近隣との交流が出来る

自治体メリット

- 人口｜共働き・第2子・第3子・若年世帯の転入定住
- 介護｜人材確保支援・重度化予防
- 安全｜通勤・飲酒運転削減・夜間交通確保
- 財政｜遊休施設の利用率を上げ、基盤を24時間活用

自治体の“課”の数だけある社会的課題に対して、人口減少を反転する生活サービス運営パッケージとして導入

顧客体験UI
Seneschal

介護・学童・夜間サービスの需要を一つの自動運転基盤で束ね、車両・運行・予約・Nodeの共用で自治体課題を解く。

https://seneschal.base44.app

Federation｜自治体共同事業で地域運行を束ねる

共同配送を起点に、医療・行政を共通運行へ接続し、地域サービスを面的に維持する



1 客室混載（制度・安全要件前提）



客室と荷室を区画・固定し、旅客・荷物双方の同乗、運行範囲、地域協議を担保

2 共同配送EV（複数荷主）



複数荷主の荷物を共同配送するEV

3 自動配送ロボット

拠点から受渡拠点迄のラスト区間を自動走行・配送



4 移動行政サービス（停車型）



停車拠点を巡回し、窓口・相談機能を提供

地域共同運行を改善するPoCループ

1. **需要・制約を把握**
   需要・ODD・交通・稼働制約
   既存サービスの利用把握
2. **人・荷物・車両を割り当てる**
   同乗・配送・車両・ロボット
   ODD・能力条件で配分
3. **周辺ODDで実証**
   人流・配送・一部ノード検証
   分担・協調を確認
4. **逸脱・希少事象を抽出する**
   介入・逸脱・再配置
   将来の拡張候補を抽出
5. **安全評価・運行評価を更新する**
   措置・安全指標・運行評価
   KPIモニタで改善
6. **承認後に運行へ段階拡張する**
   RSU・OBU／車両／拠点
   段階展開で定着

1. **認知**
   カメラ・センサー
   通知・モニター
2. **状況・リスク判定**
   ルール・不確実性
   事実確認・リスク判定
3. **行動予測**
   人・車の行動予測
   軌跡の予測
4. **相互警告**
   歩行者・車両
   通知音・介入判断
5. **希少事象抽**
   介入・逸脱・異常
   学習用付帯情報を保存
6. **仮説検証**
   シナリオ／シミュレーションODDで検証
7. **運用・ODD更新**
   運用条件アップデート
   ODDを更新
8. **地域改善・拡張**
   地域PoC→広域展開
   拠点・民間連携設計

公開技術と独自構想を、認知・学習・検証・運用更新の同じループで接続

公開情報で接続できる技術基盤

1. Woven City AI/IoT/Urban Engine・Behavior AI
   統合・運用・評価フレームによる都市実装連携
2. Woven City Integrated ANZEN System
   観測と意思決定を繋ぐ、人・モビリティ・設備・インフラの統合基盤
3. Woven City Infra Hub・Woven City Data Fabric
   都市データの統合と共有のハブ・プラットフォーム基盤
4. Automated driving・Active Learning Loop
   走行データから学習を継続し、認識精度・知識を拡張
5. Woven City Digital KAIZEN Platform
   本番利用環境における一貫した改善・検証
6. Arene SDK / Arena Tools / Arena Data
   開発フレームワーク・実行・データ・検証の運用基盤
7. Woven City xEV-VPP
   低炭素電源・シミュレーション・デジタル検証基盤
8. Woven City Robot Platform
   自動配送ロボット・移動空間ロボットの運用統合基盤

独自構想で接続できる機能

- 利用意図・需要
- 相互警告
- 都市データ連携
- 希少事象学習
- 可視化・検証
- 意思決定・最適
- エネルギー連携
- 社会実装
  設計・学習の改善ループを構成

事業・運用への接続（未実証）

- Federation｜RSU・OBU・外部データを運行制御へ接続
  地域全体での共同運行、共同配送、運行管理を実装
- 車両管理｜保守・整備・故障検知を統合管理
  整備履歴・稼働状況・給電状況を、履歴管理基盤と連携
- データ連携｜自治体・企業・住民とのAPI・Tier1
  行政、地図、医療、インフラと統合し、権限・同意を調整
- シナリオ学習｜周辺ODDから希少事象を抽出
  反復データで知識を蓄積し、適用先を強化
- Digital Fabric / Ground（基盤）
  End-to-Endデータと検証・セキュリティの基盤
- Data Up / Control Down（制御）
  収集・可視化、運用側への反映方法を設計
- PPA・Node（電源）
  水素・電力・道路・施設の連携
- 運用サービス連携（環境）
  水素・整備・充電・物流・防災などを運行基盤で統合

ラストワンマイル共同化の検証仮説

現行｜各社別配送



車種別・空車・低積載が多発
効率と交通量・事故数が悪化

構想｜地域共同配送

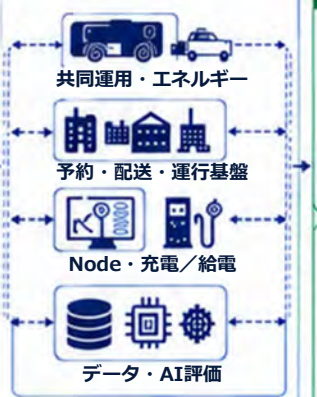


事業者別と地域横断をPoCで接続
共同化で協調・省資源・CO_2削減

1. 稼働率・事業効率を比較
2. 移動量・空車時間・エネルギーを評価
3. 1件あたりの便数・便あたり支援を検証
4. 災害時の地域配送統合化をPoCで検証

制約設定：法令制約・技術制約の下で、ラストマイル共同配送を段階運用する仮説。仮説：法令・社会要件・技術条件をPoCで検証

共同化を支える基盤



医療アクセス×自動運転｜移動・診療・薬剤配送を連携

遠隔診療・遠隔服薬指導 → 予約送迎・移動支援 → 停車中の移動診療／オンライン診療（医師・予約・ODD） → 薬局・医薬品配送管理下配送・投薬・服薬説明を記録



住民への提供価値（仮説）

- 移動料金・継続性を確保
- 診療時間・移動負担を削減

自治体が評価するKPI

- 移動のリスク・医療アクセスの改善を検証
- 医療・予約・介護連携の実効性を検証
- 災害時の医薬品配送継続性を検証

インシデント・遠隔診療の実装量と関連成功事例を活用

共同配送で車両・拠点の稼働率を高め、医療・行政サービスへのアクセス維持へ展開

事業開発環境
Federation

人流・物流・医療・行政を地域Nodeと共通運行基盤で束ね、
地域に必要なサービスを一つの自動運転ループで回す。

https://federation.base44.app

Architecture｜自治体共同事業の機能配置

自治体・運行・技術各側の役割と承認権限を分け、共同運行を継続改善できる実装構造にする

公共空間・路側領域　運行・サービス領域　クラウド・車両技術　共同基盤・合意領域　※法的責任区分ではない

CLOUD / CITY
需要・運行・検証・変更管理

公開技術：Woven City Infra Hub／Woven City Data Fabric
独自接続：需要・配車・検証・更新管理を接続

統合データ基盤

需要・予約 需要管理 / 配車・発動 最適化 運行計画 / 広域運行分析 計画 計画予約 / 統合データ基盤 / シナリオ再現 リプレイ / 安全・性能検証 SIM/実証評価 / KPI／SLA 変更管理 評価・承認 段階配信

観測データ（走行・環境・車両・V2X）／管理データ（地図・ODD・シナリオ）を分離
利用目的・同意・アクセス制御・移転履歴をデータ管理主体ごとに追跡可能。

❹ VALIDATE & RELEASE『安全検証・承認・段階配信（共同）』
モデル／地図／認可／ODDを一つの変更単位として、評価後に配信。

協働・評価データ｜走行実績・事故・車両・V2X・実証進捗の記録・ODD・監視映像

承認記録｜モデル・運行方針・地図・ODD → 変更管理ゲート

❶ OBSERVE 観測する

データ管理主体

- 道路・交通・環境
- 需要・予約・サービス
- 目的・同意・保存条件

❷ UNDERSTAND 理解する

共同

- 法令・許可・運用条件
- 災害／緊急時方針
- 地域ODD・整備計画
- 社会受容

❸ LEARN 学習する

技術提供者（オフライン）

- 各種実証環境データを統合
- 再現／検証／改善で検証
- 未承認モデルは配信しない

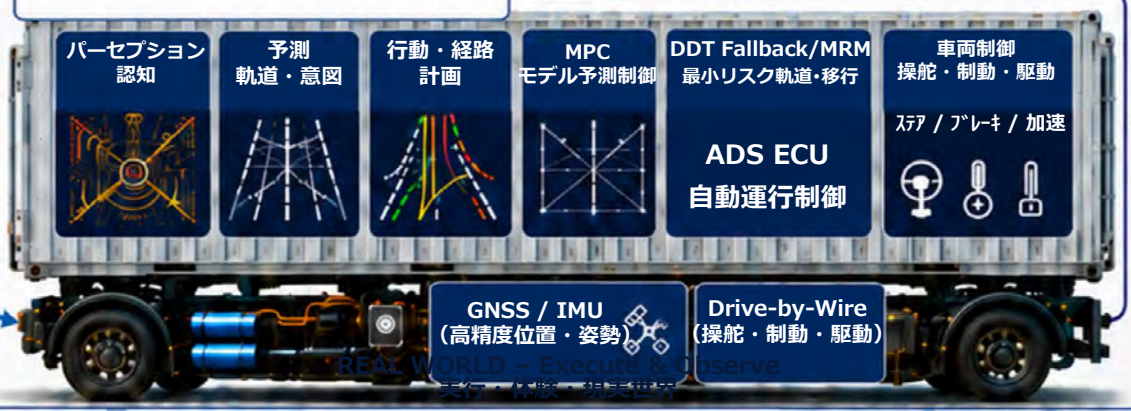
MEC / RSU – Observe & Support
観測・協調支援（車両DDTを代替しない）

路側認知 AIカメラ / センサー融合 路側センサー / 時空間統合 位置・時刻同期 / V2I / V2X C-V2X(PC5) / 局所知識生成 運行支援 / 希少事象抽出 検証対象 / RSUデータ管理 目的・権限・保存期限



RSU設備・運用
信号・カメラ・データ連携
道路管理者／警察
自治体（民間）
（合意下に運用）

OBU / VEHICLE – Decide & Execute
車載ADSが認知・判断・DDT・Fallback／MRMを担う



FC Skateboard

- 薄型低床HW（最小構成）
- 高性能FCスタック
- 低重心セーフティレイヤー
- バッテリー／冷却
- 車両コンピュータ
- スワップボディ対応

❼ OBSERVE AGAIN 再観測する

共同

- 新たな現象を観測
- 要求・動作・介入を評価
- 改善情報を抽出

❻ EVALUATE 評価する

共同

- 安全・実行評価
- 要求/インシデント検証
- 社会受容・継続判断

❺ EXECUTE 運行する

ADS＋運行事業者

- 車載ADS／DDTを実行
- 段階配信・停止・再開判断
- Fallback／MRM
- 情報開示・介入基準記録

車両／ADS機能

- ODD内でDDTを実行
- 段階配信／Fallback／MRM
- 作動状態・介入を記録
- OTA／更新管理
- サイバーセキュリティ
- 運行監視・停止判断は運行主体と分担

役割分担

- 地域要件・公共サービス：自治体
- 道路・信号・設備：道路管理者・警察
- 運行事業：運行事業者
- 車両・ADS・機能・メーカー／技術側
- KPI・監査対象・変更承認：共同

責任は法令・許可・契約に基づき個別化



DATA UP｜観測 → 学習 → 検証 → 全体へ

車両/RSU/App 観測 / AI連携 学習・希少事象 / Data Fabric 目的・権限条件 / シナリオ再生 再現・検証

- 走行・環境・車両・V2Xを分離収集
- 目的・同意・保存・公開まで追跡可能
- 品質確認後のみ学習／検証候補へ
- 生データと実運用の公開範囲を分離

CONTROL DOWN｜承認済み更新 → 段階配信

Policy/SLA 運行方針 / Model/Map 変更要求 / 安全検証/承認 変更管理 / MEC／RSU 設定作成 / OBU／車両 段階配信

- オフライン評価 → SLA／アセス → 承認 → 配信統制
- ODD整合性と安全検証・許可で配信制御
- MEC／RSUは道路管理、DDT／MRMは車載ADS
- 監査・ロールバック・インシデント対応を記録

RESPONSIBILITY｜主担当と判断原則

レイヤー	主な周期	主担当／判断
公共空間・サービス	年～月	自治体／運行主体
Cloud／Data	月～日	配信・運用・データ管理
検証／Release	月～日	検証・運行・承認主体
MEC／Edge	秒～分	設備側・即時対応
RSU／V2X	ms～秒	道路管理支援
OBU／ADS	ms	DDT・Fallback・MRM

判断者名・権限・契約・時間軸はユースケースごとに定義

自治体が地域要件、運行側がサービス・UX、技術・車両側がAI・制御・更新を担い、共同承認後に段階反映する構造。

Seneschal｜需要平準化を自動運転事業の収益性へ変える

MaaSとの支配KPI・原価の違いを整理し、24H稼働・共通運行・分散PPAを成立条件として示す

現行MaaS｜人が運転するサービス

支配KPI		支配原価	
需要平準化率	40%	人件費	45%
稼働率	30%	燃料費	40%
積載率	20%	車両維持費	10%
収益事業率	10%	その他	5%

迎車時間短縮による稼働率の向上が事業性を支配

自動運転事業｜共通基盤で24H運行

支配KPI		支配原価	
稼働率	50%	燃料費	45%
需要平準化率	25%	OS費	25%
24H稼働率	15%	車両維持費	20%
収益事業率	10%	その他	10%

人件費低減後はエネルギー費の低減が事業性を支配

燃料費の自前化を兼ねたPPAサービスを展開する事により、自動運転事業の事業基盤を構築

PPA/VPPサービス ＝ 自動運転事業の事業基盤



エネルギー運用と輸送運用を協調させ、車両・自動運転拠点・運行基盤を統合

自動運転車の配車　（自宅低圧線を時間借り）



事前予約→事前待機

突発配車→近隣迎車



迎車待ち

最短1分

自治体のメリット

遊休施設や土地を有効利用する事が出来、税金依存度を下げながら、人を呼び寄せる事が出来る

- 子育て世帯の転入・定着で人口減少を反転
- 学童・介護・物流等を統合し、行政コスト低減
- 遊休地を活用し、災害時は電源・輸送拠点へ

利用者のメリット

光熱費を下げながら、待ち時間なく、いつでも利用する事が出来、移動時間の有効活用が可能になる

- 遊休時間が95%を占める車の保有が不要
- 駐車場も削減し、好きな立寄り先で良い
- 場所を貸すだけで、エネルギー設備を購入不要

事業者のメリット

自動運転車両に必要となる拠点を構築し必要な時だけ駐車場を借りる事で固定費を下げる事が出来る

- 拠点の土地代を省き、固定費を削減
- PPAサービスによって、エネルギーコストを最小化
- 遊休時間を最適化する事で稼働率を向上

自動運転事業基盤となる拠点バリエーション

ビル拠点｜ドローンも水素充填



物流拠点｜積載中に水素充填



大型駐車場｜蓄電返却・車両清掃・水素充填



Dynamic Service（サービスの動的化）

必要な時に、必要な所へ、必要なだけ届ける

人流・移動体	自動運転	マーケティング	需給調整	セキュリティ	運用最適化・BI
需要予測	運行最適化	サービス最適化	先読み配車	不正利用防止	収益最適化

Dynamic Hub（拠点の動的化）

自動運転によって、社会のムリ・ムダ・ムラを削減

データ分析	拠点最適化	省電・発電	保守・保安	セキュリティ	統合監視・制御
稼働予測	導線最適化	OPEX	予防保全	不正侵入防止	遠隔監視

顧客体験UI
Seneschal

MaaSの個別移動最適化ではなく、稼働率・24H拠点・共通運行・分散PPAを束ね、自動運転との共用運行を持続事業に変える。

https://seneschal.base44.app

Federation｜自動運転事業成立の車両・拠点要件

積載率・24H稼働・燃料費・快適性を、FCスケートボードと自動運転拠点の設計要件へ落とす

車両要件

- 扁平化不可能なICEに代わって、電池重量・充電時間をFCで解消
- コンテナ交換による稼働率向上にて車両の減価償却費を吸収
- 自動運転に必要なバッテリー駆動による応答性を前提としたP/F
- 車載ミル水素製造にてタンク重量の問題を解消し、積載率を確保
- ミルチタン材のリソースサーキュレーションにて、最小エネルギー費
- H-伝導体MHの車載化にて、拠点での水素貯蔵・充填設備を廃止

拠点要件

- H-伝導体MHの車載化×拠点でのチタン材交換にて、相互に課題を解消
- 水道管・河川／風力ミルにて、未利用圧力を活用した電気不要な水素製造
- 回転式磁気冷凍機により、未利用圧力を活用した電気不要な水素の液体化
- 液体水素の冷熱を活用したHPC演算性能向上による付加価値向上
- 液体水素の冷熱を活用したHTSコイル巻きによる再エネ電力の貯蔵
- GTFC＋HTSモータ発電との組合せによって、一次～三次調整力に全対応

① 車載ミル（水素循環）



- 安価・オンデマンドに、走行時の回転力で水素製造
- チタン材のリサイクルとライナーの消耗抑制・長寿命化

② MS-SOFC →PC-SOFC（PCFC）



- 高効率・燃料柔軟性により、水素貯蔵の問題を解消
- 本体の小型化・出力密度の向上

③ メインシャーシー／構造体一体 H-伝導体MH



- 水素貯蔵の課題である貯蔵密度の問題を解消
- 常温・常圧で水素充填・放出可能で設備不要

④ クロスメンバー構造 HTSコイル＆LH_2タンク



- 冷熱利用によるエクセルギー効率の向上
- 応答性向上による酔わせないクルマ作り

⑤ LH_2冷却HPC（光電融合／量子）



- 悪天候時の画像処理性能向上によるLiDARレス化
- 演算・通信応答性の向上にて、E/E制御性能の向上

FCスケートボード



移動時間の有効活用を実現する為に必要な“酔わせないクルマ作り”

HTS統合車両運動制御

HTS-IWM



車両合計出力
連続300／ピーク450kW
1輪ピーク 6,000Nm以上
公称800V
作動600～900Vdc
ピーク効率 98%以上
冷却系込 50Nm/kg以上

HTSパワステ



2系統HTSモータ操舵
ラック推力 50kN以上
角度誤差 ±0.5°以内
制御周期 1ms以下
操舵応答 100ms以下
電源・巻線・制御二重化
耐久 100万km／2万h

HTSブレーキ



HTSモータ制動
20t・乾燥路・80km/h→
停止距離 50m以下
通常減速度 2.0～2.5m/s²
緊急減速度 5.0m/s²以上
指令周期 10ms以下
90%制動力 300ms以下
回生吸収エネルギー 6MJ以上／車両

HTS用SiCインバータ



4系統独立制御
連続300／ピーク450kW
ピーク時間 30秒
公称800V
ピーク効率 99%以上
出力密度 50kW/L以上
周波数 10～20kHz
耐圧 4.0kVrms・60秒

HTSエアサス×CFRP板ばね



HTS駆動アクチュエータ
制御力 ±5kN以上／輪
設計軸荷重 10t以上
制御周期 1ms以下
応答時間 50ms以下
車高 ±50mm/20s以内
鋼製比 20%軽量化
耐久 100万km／2万h

HTS-CTIS×CFRPスポーク



HTSモータ空圧制御
外径 1.1～1.2m
荷重 5～6t／輪
空気圧 600～900kPa
精度 ±20kPa
100kPa変更 2分以内
転がり抵抗 15%低減
耐久 10万km以上

進め方

① 要求分析・コンセプト設計



- ステークホルダー調整／ミッション定義整理
- ユースケース・運用設計・CAE/CFD解析
- 要求分析・トレードオフ・技術検証課題
- システム構想・実証アーキテクチャ定義

② 要素技術開発・設計



- システムアーキテクチャ／材質・構造設計
- システムモデル化・構造／衝突安全設計
- シミュレーション／HMI／制御探索
- 評価計画・試験計画・試作設計調整

③ E置拠点向けで立上げ



- 実証運用拠点による本番型実証
- 性能評価・改善効果検証
- 安全性・環境性・拠点最適化
- 保安・防錆・運用ルール整備

④ 五月雨式に車載化



- 耐振／耐久評価・衝突安全・法規適合
- コンテナ運用・運用台数・保全の最適化
- 連携体制・コスト最適化
- 導入先のサービス＆オペレーション整備

HT-PEFCによる本体小型化効果と比較検証

サービス車両に求められる少量生産・最新技術の取込みに適した作り方

事業開発環境 Federation

積載率・24H稼働・燃料費・快適性という事業成立条件を、FCスケートボードと自動運転拠点の具体構造へ落とし込む。

https://federation.base44.app

Architecture｜自治体実装を支えるエネルギー・演算基盤

車両・拠点・都市をエネルギー・演算・通信でつなぎ、街事業を支える共通基盤へ拡張する

交通・物流データ
大規模データを
リアルタイムに処理

再生可能エネルギー基盤
再エネ電力調整力・水素輸送・
極低温供給

分散HPC基盤
極低温を活用した
AI演算・冷却

**車両・都市間の
エネルギー最適化**
電気・水素の需給を
車両・都市間で最適化

社会システム融合
街と一体となった
自動運転モビリティ

住宅地 Node
戸建て住宅の駐車場で水素充填しながら待機または近隣から迎車する事により、クルマ利用へ乗り換える心理的ハードルとなる迎車時間の課題を解消。
（他社にやられる前に自らが事業転換）

物流拠点・大型駐車場 Node
物流拠点：バースでの荷役中 or 商業施設：お客様の買い物中に水素充填する事により、エネルギー補充の待ち時間を実質ゼロとし、コンテナ交換と合わせて、MaaSの事業性確保で求められる稼働率を最大化。
（蓄電とのハイブリッドでエネルギーを安定供給）

ビル Node
ペロブスカイトPVで発電された電力を貯蔵し、再生可能エネルギーを高速需給調整可能なHTSケーブル巻きLH2パイプラインによって、分散型HPCのパフォーマンスも同時に向上
（自動運転事業の屋台骨を手の内化）



街事業開発を進める"Node Mesh"



光・量子ネットワーク × 分散コンピューティング基盤

古典演算・通信
- 既存接続
- 汎用制御

オールフォトニクス
- 低遅延
- 大容量
- 省電力

量子演算・通信
- 高秘匿
- 複雑最適化
- 高精度同期

RSU・OBU 分散コンピューティング



分散演算のメリット（Edge）
- 低遅延：リアルタイム制御
- 高信頼：クラウド等統合制御
- 協調制御をローカル分散で継続
- 用途分野を横断し全体へ還元

制御対象：モビリティ中心のエネルギーとアクティビティ

運行制御
加速・減速・停止
車線制御・信号制御

エネルギー制御
蓄電・放電・
補給・充電状況
・冷熱制御

充電・充填
給油・充電・充填
最適化・コンテナ
交換・最適運用

輸送・配送
温度要求・
充電最適化
冷蔵・冷凍最適化

サービス連携
乗客・荷物・
支援連携・施設・
決済マッチング

都市アクティビティ
人流・物流
・交通・街の
GIS連携・最適化

ユースケース



分散・低遅延・自律最適の都市モビリティ基盤



法規・制度
法整備・制度化

標準・仕様
相互運用・規格化

資源調達・社会実装支援
パートナー連携
導入・初期運用支援

金融・投資
資金調達・投融資支援

技術・研究
データ基盤・運営

ビジネス／サービスで事業基盤につなげる連携
付加価値の創出・多様化

地域連携・ガバナンス
住民参加・合意形成

Overview

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Autono-MaaS Business Logic

Seneschal: https://seneschal.base44.app ・・・ Demand-generation implementation example
Federation: https://federation.base44.app ・・・ Urban-operation packaging study framework

The concepts presented in this portfolio are not proposals for implementation. They demonstrate my concrete understanding of the strategy and execution requirements needed to make autonomous-mobility businesses commercially viable.

The success of Autono-MaaS is determined not by autonomous driving technology itself, but by a business structure that converts operation into profit. As driver labor cost decreases through automation, the dominant cost factor converges toward energy cost, while vehicle utilization is required to recover CAPEX and load factor is required to increase revenue per trip. Seneschal smooths demand fluctuations by matching customer availability with mobility and services, while motion-sickness-free vehicle control transforms travel time into work, rest, and service value.

Because vehicle-grade quality requirements are strict, next-generation vehicle technologies should first be developed for stationary applications, then installed into service vehicles as they become ready. Service vehicles can absorb CAPEX through high utilization and high load factor, enabling a DevOps-style development model unlike conventional mass-production vehicles. Federation increases vehicle utilization through time-based service exchange and ridesharing, and increases load factor through passenger-freight integration and collaborative logistics, while planned mobility schedules optimize PPA energy management and enable AVP-integrated fueling to establish a carbon-neutral energy platform for autonomous mobility. Autono-MaaS is an urban operating system that controls demand, vehicles, energy, and infrastructure as one system to convert utilization, load factor, and energy cost into profit.

PPA Service for Autonomous Mobility Foundation

As autonomous driving reduces driver labor cost, energy becomes the dominant operating-cost factor. The PPA service therefore establishes structural energy-cost competitiveness for Autono-MaaS through distributed energy infrastructure.

Corporate PPA services support Scope 3 decarbonization while establishing distributed eVTOL operating hubs. Idle corporate vehicles are shared for employee commuting to improve fleet utilization and reduce Scope 3 transportation emissions. Residential trigeneration services establish distributed neighborhood mobility hubs, providing the node density required for daily autonomous mobility services.

Blockchain-enabled P2P energy and mobility sharing connects corporate and residential nodes into a shared mobility and energy ecosystem, while renewable electricity, water-pipeline micro-hydropower, magnetic refrigeration, hydrogen production, superconducting power cables, liquid-hydrogen pipelines, and liquid-hydrogen-cooled HPC establish the infrastructure foundation for infrastructure-coordinated autonomy and continuous ODD expansion.

Infrastructure Layer	Control Variable	Target	Business Purpose
Renewable PPA	Renewable supply ratio	Maximize	Low-cost renewable electricity
Water Infrastructure	Micro-hydropower output	Maximize	Distributed baseload generation
Hydrogen Production	Electrolysis utilization	Maximize	Hydrogen production
Cooling	Magnetic refrigeration efficiency	Maximize	LH_2 production & HPC cooling
Energy Network	Superconducting cable utilization	≥80%	Inter-node power sharing
Hydrogen Network	LH_2 pipeline utilization	≥70%	Hydrogen distribution
HPC Node	AI compute availability	≥99.9%	Infrastructure AI computing
Infrastructure Coordination	Infrastructure coverage	Expand	Vehicle–infrastructure cooperation
Autonomous Mobility	ODD coverage	Expand	City-wide autonomous operation

AVP-to-MaaS Execution

This concept begins with an existing OEM customer use case: enabling refueling to be completed while the user is shopping. → A: time control

By integrating infrastructure-coordinated automated valet parking (AVP) with fueling, vehicles autonomously park, refuel, and return within dwell time, eliminating dedicated refueling stops. → B: time + location control

As autonomous driving progresses, parking demand shifts, creating underutilized parking spaces repurposed as operational nodes. By leveraging these assets, dual-use infrastructure enables daytime refueling for private users and nighttime fleet operations. → C: time allocation control

Liquid-hydrogen-cooled HPC nodes provide compute capacity for infrastructure-coordinated autonomy. Any ODD expansion remains subject to scenario coverage, V&V evidence, a validated safety case, and regulatory approval.

Applied to large-scale parking operators, operational nodes exchange modular service containers according to time-of-day demand, balancing utilization across mobility, logistics, retail, and public services. This transforms Node into a MaaS foundation that mitigates demand fluctuations and increases asset utilization. → D: operational control

Extending this logic to public roads evolves the system into a city-wide mobility platform. → E: flow control

Through autonomous driving, vehicles synchronize in time and space, forming organized carrier operations rather than independent movements. → F/G: synchronization / platoon control

This expansion to public roads and coordinated carrier systems enables a new transportation system. Because this system is built on coordinated flows and shared infrastructure, Autonomous mobility expands from vehicle operation into integrated mobility and infrastructure coordination. → H: system-level control

Logic	Layer	KPI	Formula	Value	Unit
A	Demand	Demand generation rate	generated demand ÷ total potential demand	≥0.30	ratio
A	Demand	Conversion rate	booking ÷ request	≥0.23	ratio
B	Demand	Matching rate	matched ÷ request	≥0.90	ratio
D	Demand	Trip request rate	executed trips ÷ matched	≥0.95	ratio
D	Operation	Dispatch success rate	completed dispatch ÷ dispatch request	≥0.95	ratio
D	Operation	Dispatch response time	total dispatch response time ÷ dispatches	≤60	sec
D	Operation	Utilization	operating time ÷ total time	≥0.75	ratio
D	Operation	Fleet idle ratio	idle vehicles ÷ total vehicles	≤0.25	ratio
F	Vehicle	Latency	response time	≤1	sec
F	Vehicle	Control delay	actuation delay	≤20	ms
F	Vehicle	Planning latency	planning delay	≤100	ms
G	Vehicle	Control stability	spacing variance	≤5	%
E	Infrastructure	Throughput	vehicles ÷ time	≥40	veh/h
C	Infrastructure	Node utilization	used nodes ÷ total nodes	≥0.80	ratio
F	Infrastructure	Node latency	node delay	≤50	ms
H	Infrastructure	Energy cost per trip	energy cost ÷ trips	≤¥500	JPY/trip

PORTFOLIO

Autonomous Mobility Business Execution Architecture

— Driving autonomous mobility business development through AD/ADAS system foundation —

Autonomous Mobility Business Definition

This architecture is defined as a control system where KPI, system behavior, and business outcomes are directly connected. This architecture directly connects KPI, system behavior, and business outcomes as a single control system. The objective is to build an AI-enabled autonomous mobility business architecture that improves utilization, load factor, and unit economics.

This is achieved by integrating mobility, logistics, energy, and urban services through a common operating architecture, while structurally improving:

- utilization
- load factor
- unit economics

This proposal defines a business architecture that makes this system executable, economically grounded, and deployable.

Autonomous Mobility Business Execution

This proposal defines a phased development and deployment plan for autonomous mobility business execution. Today's mobility systems fail to scale because demand, fleet, energy, and operations are still managed separately. This architecture integrates these elements into a unified autonomous mobility business structure, using autonomy as the foundation and AI as the means to operate autonomous mobility at business scale. System performance directly translates into revenue through utilization, load factor, and energy efficiency:

• Demand creation synchronized with dispatch
• Utilization maximization through modular vehicle operation
• Load factor optimization through dynamic vehicle composition
• Energy cost reduction through on-site hydrogen production
• Real-time operational control through distributed node coordination and AI-enabled optimization

Autonomous Mobility Business Architecture

Objective: To define an executable autonomous mobility business structure where AI-driven control logic is applied through autonomous driving to integrate demand, operations, vehicles, and infrastructure.

This architecture shifts mobility from fragmented services into a unified operating system where utilization, load factor, and energy efficiency are structurally optimized.

AI is applied as controlled logic across four layers.

1. Demand Orchestration

2030: Coordinated AI Assistant
A calendar-based interface connects user schedules with service availability, enabling demand generation through matching and reservation-linked mobility.

2035: Trusted AI Partner
A 4D digital platform integrated with map infrastructure synchronizes city-state and user behavior in real time, enabling continuous demand generation and capture to maximize load factor.

2. Operation Control

2030: Integrated Control
Energy management and dispatch are coordinated through decision logic that can later be augmented by AI, enabling stable and efficient operations under constrained conditions.

2035: Synchronized Control
Real-time synchronization of in-car and out-car hydrogen and electricity flows creates the control foundation for AI-assisted optimization, while structurally improving business viability.

3. Fleet Vehicle

2030: Swap Carrier
A modular vehicle structure enables continuous operation through container exchange, maximizing utilization.

2035: Composed Carrier
Vehicles dynamically couple and reconfigure during operation, enabling real-time redistribution of passengers and cargo to maximize load factor.

4. Distributed Infrastructure

2030: Urban Node
Distributed hubs provide localized energy and compute support, enabling operational independence and cost efficiency.

2035: Node Mesh
Distributed infrastructure is synchronized as a network, enabling real-time coordination between vehicles, energy, and computation.

This architecture enables a system where mobility, energy, computing, and AI-based decision-making operate as a unified layer, enabling scalable autonomous mobility business operation.

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AI-Enabled Urban Mobility Roadmap

Autonomous mobility becomes deployable only when a production-ready driving stack, closed-loop learning, auditable safety, fleet operations and service economics are developed as one system. This roadmap first establishes that industrial baseline, then extends AI control from vehicles to distributed infrastructure and city-scale operations.

Industry Baseline & Capability Build-Up

Capability	Industry Baseline	Roadmap Focus
Product / Release Governance	Common requirements, configuration control, release evidence and change management across vehicles, sites and ODDs	Stage deployment through safety, service-readiness, availability, recovery-time and unit-economics gates.
Driver / Sensor Platform	Multimodal camera–lidar–radar perception, overlapping coverage, redundancy and deployment across multiple vehicle classes.	Common driving stack, hardware abstraction, calibration, fail-operational design and rapid vehicle onboarding.
Data / Active Learning	Fleet-scale data collection, anonymization, annotation, rare-event mining and uncertainty-based active learning.	Closed loop: detect → anonymize → select → label → train → validate → staged release.
Simulation / V&V	Real-event replay, scenario variation, closed-course and public-road tests, synthetic simulation and automated regression.	ODD-specific scenario coverage, acceptance criteria, critical-event recall, safety evidence and validation gates.
Safety / Remote Support	ODD safety cases, independent evidence, fallback/MRM, remote monitoring and contextual assistance with clear authority boundaries.	Claims, evidence, escalation, incident response, cybersecurity, rollback, audit trail and remote-support limits.
Fleet / Service Operations	Integrated dispatch, maintenance, charging, incident management, customer support and feedback from daily autonomous operations.	Connect the driving stack with fleet and service control; manage SLA, availability, intervention, energy and cost/hour.
Scale / Economics	Reusable software/hardware platforms, lower-cost sensors and compute, standardized integration and multi-vehicle deployment.	Track utilization, launch lead time, ODD rollout speed and lifecycle revenue; separate common and use-case-specific functions.

The industrial baseline combines a production-ready driver, closed-loop learning, scenario-based V&V, auditable safety, remote support, fleet operations and repeatable ODD deployment. The roadmap establishes these capabilities first, then differentiates through city data, infrastructure coordination, demand orchestration and business optimization.

AI-Controlled Mobility System

This section defines how AI functions as control logic across layers and is executed through autonomous driving systems. AI is applied differently at each layer — from demand prediction to system-wide coordination — but all functions are integrated through a unified control structure that enables real-time execution at city scale.

Layer	Cloud	Distributed Infrastructure	Fleet Vehicle
Perception	2030: Env-state estimation (fusion, uncertainty propagation) / 2035: City-wide world-state estimation (global sync, uncertainty minimization)	2030: Roadside perception augmentation (edge fusion, low-latency inference) / 2035: Cooperative perception (shared state, spatiotemporal fusion)	2030: Onboard perception (LiDAR / Camera / Radar, ≤10ms) / 2035: Infra-coordinated perception (shared scene state)
Planning	lispatch optimization (forecast, :onstrained search) / 2035: City nission optimization (global :cheduling, real-time recompute)	2030: Zone operation optimization (routing / berth / node) / 2035: Real-time replanning (node sync, global constraints)	2030: Local path / motion planning / 2035: Cooperative trajectory planning
Control	2030: Learning-based control (constraints, stability) / 2035: Global closed-loop optimization (adaptive)	2030: Edge feedback control (low-latency, constraints) / 2035: Distributed synchronized control (coordination, load)	2030: Motion control (MPC, ≤50ms, constraint-aware) / 2035: High-frequency adaptive control (≤10ms, cooperative)
Coordination	2030: Fleet-state coordination (dispatch-state sync) / 2035: Fleet-wide coordination (global sync, decision integration)	2030: Node-state coordination (local sync) / 2035: Node Mesh synchronization	2030: Vehicle-level coordination / 2035: Platoon coordination

ODD / DDT as Revenue Driver

ODD / DDT transition defines idle time and directly determines utilization and revenue structure.

DDT Loss = (1-TSR) x Recovery time + TOT, Revenue (OAR x UAC) x (1÷(Trip+(1-TSR) x Recovery + TOT) x FLC x Load

Execution Sequence & Exit Gates

The roadmap advances only when technical evidence, operational readiness and unit economics are verified together.

Phase	Capability Build	Exit Gate	Business Outcome
1. Industrialize	Common driving and sensor stack; closed-loop data; scenario V&V; auditable safety case; remote operations and field support.	Repeatable launch within the initial ODD, with controlled fallback, rollback, incident response and configuration management.	Production service with measurable safety, availability, intervention rate, recovery time and cost/hour.
2. Federate	Fleet orchestration; RSU / OBU coordination; shared data and simulation; distributed infrastructure and service control.	Cross-vehicle and cross-site operation under common interfaces, authority boundaries, release procedures and KPI governance.	Lower intervention, recovery time and idle capacity; faster vehicle, site and ODD onboarding.
3. Scale	4D demand orchestration; Node Mesh; energy-compute integration; evidence-led quantum computing, sensing and optimization.	Multi-ODD, multi-service and multi-city performance with resilience, traceability and viable unit economics.	Scalable urban mobility platform with recurring service, infrastructure and lifecycle revenue.

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Urban AI Operating Architecture

An integrated capability progression from real-world understanding and coordinated safety to governed learning and city-scale operations.

1. UNDERSTAND & PROTECT Real-world intelligence and coordinated safety	2. CONNECT & IMPROVE Governed data, validation and continuous learning	3. OPERATE & SCALE Distributed execution and city-scale orchestration
AI Vision Engine — Builds a multimodal understanding of people, vehicles, infrastructure, road conditions and the surrounding urban environment.	Infra Hub — Connects roadside, mobility, building and public-service infrastructure through secure, governed and interoperable interfaces.	4D demand orchestration — Predicts where, when and why demand will emerge across mobility, logistics, energy and public services.
Behavior AI — Predicts human and vehicle intent, interaction patterns, anomalies and uncertainty to support context-aware decisions.	Data Fabric — Federates vehicle, city, infrastructure and operational data without forcing every participant into a single centralized system.	Rare-event learning — Prioritizes edge cases, near misses and low-frequency, high-impact events for targeted learning and validation.
Drive Sync Assist — Synchronizes driving support, shared awareness and timely intervention across the driver, vehicle and surrounding environment.	Digital KAIZEN — Converts operational evidence into simulation, validation, controlled deployment and measurable continuous improvement.	RSU / OBU coordination — Distributes perception, routing, safety and operational tasks across roadside and onboard systems.
Integrated ANZEN — Integrates detection, prediction, risk assessment and intervention, including graduated assistance, fallback and minimum-risk maneuvers.	Scenario simulation and validation — Tests policies, ODD changes, service patterns, traffic conditions and rare events before field deployment.	Node Mesh synchronization — Coordinates vehicles, roadside nodes and local service hubs as a resilient, distributed operating network.
Multimodal real-world understanding — Fuses cameras, sensors, maps, V2X signals and contextual data into a consistent operational picture.	Operational feedback loop — Captures outcomes, deviations, user behavior and system performance to update models, rules and operating procedures.	Energy-compute integration — Allocates charging, storage, edge computing and fleet tasks against demand, cost, capacity and resilience.
Predictive risk and uncertainty management — Identifies emerging conflicts early enough to preserve human decision time and maintain a safe operating margin.	MBD / digital-twin lifecycle loop — Links design assumptions, simulation results, field evidence and software or policy updates throughout the lifecycle.	Quantum computing and sensing — Explores optimization and sensing gains for high-dimensional mobility, infrastructure and urban-service operations.
Coordinated safety — Aligns human handover, vehicle fallback, infrastructure assistance and accountable decision-making across the operating environment.	Governed continuous improvement — Releases only validated updates with traceability, approval, rollback capability and measurable safety evidence.	Quantum-enhanced perception — Applies advanced sensing and inference only where they demonstrate meaningful gains over classical approaches.
Shared scene awareness — Makes relevant hazards, intent and operating context understandable across people, vehicles, infrastructure and service operators.	City-data integration — Connects mobility, energy, logistics and public-service operations through governed, reusable data products.	Full-city synchronization — Coordinates cross-domain actions while preserving local autonomy, safety boundaries and service continuity.
Safety-to-ODD expansion — Extends system capability only after evidence demonstrates acceptable risk and operational readiness in each target condition.	Closed-loop execution — Deploys approved models, policies and operating rules back to vehicles, RSUs, infrastructure and service operations.	Autonomous city-state optimization — Continuously balances safety, service quality, capacity, energy consumption, resilience and accessibility.
Human trust and adoption — Explains system state, preserves appropriate human authority and captures acceptance signals from users and operators.	Governance, KPI and business control — Monitors safety, availability, unit economics, energy, inclusion, service quality and lifecycle performance.	Scalable business execution — Converts technical gains into repeatable operations, partnerships, asset management and sustainable lifecycle value.

Cross-Layer Decision Rights & Release Gates

Control Domain	Primary Authority	Evidence Returned Upward	Release / Deployment Gate
Vehicle / Edge	Immediate perception, motion control, fallback, minimum-risk manoeuvre and local safety intervention.	Uncertainty, hazard traces, interventions, sensor and actuator health, and safe-state results.	Latency, hazard-response coverage, fail-operational capability and safe-state completion.
Fleet / Infrastructure	Dispatch, routing, RSU / OBU coordination, remote assistance, node control and energy-compute allocation.	Availability, dwell and recovery time, node congestion, service exceptions, energy use and capacity.	SLA, intervention rate, recovery time, operational continuity and ODD-support readiness.
Cloud / City	Model and policy release, scenario validation, governance, cross-domain learning and city-scale optimization.	Scenario coverage, causal evidence, audit trail, service quality, cost and lifecycle-performance KPIs.	Traceability, rollback readiness, validated KPI improvement, resilience and sustainable unit economics.

Validated policies, models and operating rules move downward; evidence, uncertainty, incidents and business KPIs move upward. Safety authority remains local, while learning and optimization remain globally governed, measurable and auditable.

Across these three layers, the operating cycle remains Observe → Understand → Learn → Evaluate → Execute → Observe Again. Real-world signals are interpreted and protected at the edge; governed data is federated and converted into validated improvements; approved policies and models are then deployed back to vehicles, RSUs, infrastructure and urban operations. Evidence and learning move upward, while only validated policies and control updates move downward. The architecture scales from coordinated assistance to cross-domain orchestration across mobility, logistics, energy and public services, while preserving human accountability, fallback and rollback. Quantum computing, sensing and quantum-enhanced perception are introduced as evidence-led accelerators—not assumptions—and must demonstrate measurable gains in safety, resilience, service quality, capacity or lifecycle economics before deployment. The result is a practical path from isolated AI functions to a trusted, continuously improving urban operating system.

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Layer-integrated Control & Business Architecture

This architecture defines how business requirements are translated into executable system design through AI-driven control and autonomous driving.
It connects KPI targets with technical execution, ensuring that system performance directly drives economic outcomes.

Executable mobility operation

This section maps business KPIs to control logic and AI-relevant technical conditions required to achieve them.

	2030			2035		
	Logic	**Input**	**Target**	**Logic**	**Input**	**Target**
Perception	Match ≤500ms [A] Update ≤5min Trigger ≤1s Conv ≤1min Err ≤20% Succ ≥85%	Cal ±5min Res ≤1s Avail ≤500ms	CVR ≥23% Wait ≤6min Cancel ≤12% Eff conv≥19.6% Match ≤500ms Err ≤20% Trigger ≤1s	Match ≤100ms [A] Update ≤1s Trig ≤100ms Conv ≤10s Err ≤5% Reflect ≥95%	Behavior ±1s History ≤10s Avail ≤100ms	CVR ≥35% Wait ≤3min Cancel ≤5% Eff ≥33.3% Matc ≤100ms Err ≤5% Trig ≤100ms
Planning	Disp ≤1min [B] Route ≤5s Alloc ≤10min Loop ≤5min Succ ≥95%	Fleet ±50m ETA err ≤3min Traffic ≤30s	Trips ≥73/d Loaded ≥14/d Resp ≤8min ETA err ≤3min Empty ≤25% Util ≥75% Route ≤5s	Disp ≤10s [B] Route ≤1s Alloc ≤1min Loop ≤1min E-int ≥95% Succ ≥99%	Fleet ±5m Traffic ±1s Sync ≤5s	Trips ≥140/d Load ≥112/d Resp ≤5min ETA err ≤2min Empty ≤15% Util ≥85% Route ≤1s
Control	MPC ≤50ms [C] Traj ≤50ms Conv ≤100ms Stability ±15% Fail-safe ≤100ms	Percep ≥95% Plan ≤200ms V2X ≤20ms	Price ≥¥2,200 Ctrl ≤50ms Plan ≤200ms Err ≤30cm Intv ≤1/100km Util ≥87.5%	MPC ≤10ms [C] Traj ≤10ms Conv ≤10ms Stability ±5% Coop ≥95% Fail-safe ≤10ms	Perce ≥99.9% Plan ≤50ms Sync ≤10ms	Price ≥¥2,000 Ctrl ≤10ms Plan ≤50ms Err ≤5cm Intv ≤1/10,000km Util ≥95.8%
oordination	Alloc ≤1min Compute ≤1s Refuel ≤5min Conv ≤10min S-D ≥0.78 Load err ≤10%	Energy ±5% [D] Comp ≤100ms Sync ≤1min	OPEX bound Comp ≤150ms Eff ≥75% Util ≥78% Eff energy ≥58.5% Refuel ≤2/d Var ≤±5%	Alloc ≤100ms Comp ≤10ms Refuel ≤1min Strategy ≤1h Conv ≤1min S-D ≥0.96 Load err ≤3%	Energy ±1% [D] Comp ≤10ms Sync ≤10s	OPEX≈¥202.8K/d Comp ≤50ms Eff ≥90% Util ≥96% Eff energy ≥86.4% Refuel ≤1/d Var ≤±1%

Operating index (dimensionless) = Utilization × Load Factor × Energy Efficiency; profit still requires price, volume, and cost.
Available service time = scheduled time − [(1−TSR)×recovery time + TOT]; Revenue = Trips × Seats × Load × Fare.

Layer-Integrated Business Architecture

This section defines business requirements and constraints., and translates them into executable system architecture.
AI is applied as controlled logic across four layers.

	2030		2035	
	Technology	**KPI**	**Technology**	**KPI**
Service Orchestration	・Calendar-driven Demand Orchestration Engine ・Service Reservation ・Availability Matching Engine ・Reservation-triggered Dispatch ・Linkage API ・Customer Touchpoint PaaS ・Context-aware Notification	CVR ≥23% Wait ≤6min Cancel ≤12% Match ≤500ms [A] Avail ≤500ms Res ≤1s Cal ±5min Trigger ≤1s	・Behavioral AI Orchestration ・Agent-Demand Control ・4D Digital Twin ・Travel-time Monetization ・City Service Optimization ・Fleet Camera social sensing ・IFTTT Trigger Dispatch ・Autonomous Intent Agent	CVR ≥35% Wait ≤3min Cancel ≤5% Match≤100ms [A] Avail ≤100ms Behavior ±1s History ≤10s Reflect ≥98%
Operation Control	・Dispatch Optimization (VRP) ・Real-time Routing ・Closed-loop Dispatch ・Demand Prediction ・Energy-aware Routing ・Demand-Supply Balancing ・ETA Prediction ・Capacity Allocation ・ETA Feedback Control ・Queue Optimization	Trips ≥73/d [B] Load ≥14/d Resp ≤8min ETA err ≤3min Route ≤5s Alloc ≤10mi Fleet ±50m Traffic ≤30s Empty ≤25% Util ≥75%	・Integrated TMS ・Fleet-cooperative Control ・Energy-Transport Optimizer ・Quantum-inspired Optimization ・In / Out Energy Sync ・Real-time Replanning ・Constraint-aware Optimization ・Multi-agent Coordination ・Global State Synchronization ・Predictive Dispatch	Trips ≥140/d [B] Load ≥112/d Resp ≤5min ETA err ≤2min Route ≤1s Alloc ≤10s Fleet ±5m Traffic ±1s Empty ≤15% Util ≥85%
Fleet Vehicle	・Swappable Vehicle Module ・MS-SOFC＋SSB ・LH2 Tank TypeⅣ Subframe ・MgH2 MH Main chassis ・PMSM Distributed Drive ・SiC Inverter ・SbW (Dual Motor Redundant) ・BbW (Electro-Hydraulic +Regen) ・CFRP Leaf + Active Suspension ・CFRP Spoke + CTIS Tire	Util ≥87.5% [C] Err ≤30cm MPC ≤50ms Traj ≤50ms Conv ≤100ms Stability ±15% Fail-safe ≤100ms Percep ≥95% Plan ≤200ms Intv ≤1/100km	・Dynamic Vehicle Coupling ・Onboard Mill+ Al Air Bat ・HTS coil winded LH2 Tank ・H- Conductor adapted MH ・Real-time energy optimization ・Thermal-compute coupling ・High-frequency power control ・Quantum-enhanced LiDAR ・Quantum magnetometer ・Atonic clock	Util ≥95.8% [C] Err ≤5cm MPC ≤10ms Traj ≤10ms Conv ≤10ms Stability ±5% Coop ≥98% Fail-safe ≤10ms Perce ≥99.9% Plan ≤50ms Intv ≤1/10,000km
Distributed Infrastructure	・High-precision docking ・Wind-driven Mill ・Wind-driven AMR ・SOFC + Ammonia Tank ・Superconducting Power Cable ・6G Communication Network ・Standardized energy interface ・Low-latency communication ・Sub-ms time synchronization	Eff ≥75% Util ≥78% Energy ±5% [D] Compute ≤1s Sync ≤1min Refuel ≤5min S-D ≥0.78 Load err ≤10% Var ≤±5%	・Precision docking ・Ultra-low latency V2X sync ・Water-pipeline Mill ・Pipeline-integrated AMR ・PCFC ・H-conductive BIMH Storage ・HTS-coiled LH2 Pipeline ・Quantum Communication ・Energy-transport interface	Eff ≥90% Util ≥96% Energy ±1% [D] Compute ≤10ms Sync ≤1s Refuel ≤1min S-D ≥0.96 Load err ≤3% Var ≤±1%

I design systems where business requirements and technical execution are structurally aligned.

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System Design for Mobility Feasibility

This page defines scenario design targets for Energy Efficiency = 0.58 (=58%), Utilization = 0.75 (=18h/day ÷ 24h/day), and Load Factor = 0.70 (=2.8pax ÷ 4pax). They are not proven outcomes; each target requires metered validation through integrated system architecture.

The design is structured across energy supply [kWh/day], vehicle dynamics and response [ms, cm, Hz], and control and computation [ms, Hz]. Net traction demand is 2,080.5kWh/day (=28.5kWh/trip × 73trips/day). At 60% electrical efficiency, required fuel-LHV input is 3,467.5kWh/day and the hydrogen-equivalent requirement is 104.1kg/day (=3,467.5 ÷ 33.3). Annual PV energy can exceed annual net demand, but an annual ratio does not prove 24-hour supply without power, storage, loss, and weather modelling.

Vehicle and control design ensure "motion sickness free" operation, defined by control latency ≤20ms, decision latency ≤100ms, positioning error ≤10cm, and vibration ≤0.3Hz. These are maintained through superconductivity enabled by liquid hydrogen cooling and real-time control via HPC operating at 100Hz (=1 ÷ 0.01sec). The HPC value is a design target and does not by itself prove continuous satisfaction or ODD expansion.

Block	Variable	Corrected value	Formula / definition	KPI link
KPI	Energy Efficiency	0.58 target	Σ useful traction output ÷ Σ purchased energy input	Energy
	Utilization	0.75 (=18÷24)	18h/day ÷ 24h/day	Operation
	Load Factor	0.70 (=2.8÷4)	2.8pax ÷ 4pax	Demand
Constraint	Operation Hours	24h/day	operating-window target	Utilization
	Stability	0.95	1 − 0.05	Energy
	Throughput	40veh/h	60min ÷ 1.5min	Infra
Energy	Net traction demand	2,080.5kWh/day	28.5×73	Energy
	Fuel-LHV input	3,467.5kWh/day	2,080.5÷0.60	Energy
	SOFC electrical efficiency	0.60	net electricity ÷ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	source constant	Energy
	H2-equivalent requirement	104.1kg/day	3,467.5÷33.3	Supply
	Annual net demand	759,383kWh/yr	2,080.5×365	Energy
	PV capacity	1,500kW	1.5MW×1,000	Energy
	PV yield	1,100kWh/kW/yr	scenario benchmark	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	1,500×1,100×0.85	Energy
	Annual PV ratio	1.85	1,402,500÷759,382.5	Annual energy only
	24h supply gate	Not proven by annual ratio	hourly power + storage + loss + weather model	Infra
	Supply loss	0.05 target	5%	Energy
	Supply stability	0.95 target	supplied ÷ required energy	Energy
Node	Node Delay	≤50ms	measured end-to-end node delay	Infra
	Dispatch Interval	5sec	60sec ÷ 12	Utilization
	Trips	73/day target	scenario target; capacity ceiling 40×12=480/day	Demand
	Fleet	13 vehicles	ceil(73÷6)	Operation
	Load capacity	4pax	2×2 seats	Demand
	Energy per trip	28.5kWh/trip	2,080.5÷73	Energy
	Continuous ratio	1.00 target	24÷24	Utilization
	Availability	0.99 target	available time ÷ scheduled time	KPI
	Reliability	0.99 target	successful cycles ÷ total cycles	Stability
Vehicle	Control Sync	≤20ms	100ms ÷ 5	Motion sickness free
	Decision Sync	≤100ms	100ms	Motion sickness free
	Position Error	≤10cm	1m ÷ 10	Motion sickness free
	Platoon Stability	≤5%	0.05	Motion sickness free
	Control Cycle	100ms	1sec ÷ 10	Stability
	Response Delay	≤20ms	100ms ÷ 5	Motion sickness free
	Accel Jerk	≤0.5m/s²/s	1 ÷ 2	Motion sickness free
	Vibration	≤0.3Hz	1Hz ÷ 3.33	Motion sickness free
	Sync Accuracy	0.99	1 − 0.01	Stability
Control	Prediction Rate	1.00 target	successful predictions ÷ attempts	Operation
	Planning Rate	1.00 target	successful plans ÷ attempts	Operation
	Execution Rate	100Hz	1 ÷ 0.01sec	Control
	HPC Latency	≤10ms	measured inference latency	Stability
	Node Sync	≤50ms	measured synchronization delay	Stability
	Flow Capacity	40veh/h	60÷1.5	Infra
UX	Utilization Gain	0.10 target	0.75−0.65	KPI
Comparator	PEMFC efficiency	~60% direct H2	DOE technology comparison; not 22%	No exclusion on efficiency alone
	Battery downtime	4h/day assumption	24−20	Scenario only
	Battery utilization	0.83 assumption	20÷24	Scenario only
	ICE efficiency	0.30 assumption	scenario benchmark	Scenario only
Physics	Annual energy ratio	1.85	PV annual generation ÷ annual net demand	Energy
	Continuous supply	Validation pending	hourly balance required	Utilization
	Conversion loss	0.40	1−0.60	Energy
System	Feasibility gate	Not proven by multiplication	all independent safety, energy, operation, and financial gates must pass	Final
Conclusion	Hydrogen equivalent	104.1kg/day	2,080.5÷0.60÷33.3	Core
	Node operation	Validation pending	24h hourly model required	Core
	Utilization	0.75 target	18÷24	Core
	KPI status	Validation pending	not 0.58×0.75×0.70	Final

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Feasibility Conditions for Autonomous Mobility System

This page does not present financial outcomes. It defines scenario conditions that must be independently validated. Energy Efficiency = 0.58, Utilization = 0.75, and Load Factor = 0.70 are design targets, not derived facts. Feasibility depends on energy supply [kWh/day], available power [kW], storage duration [h], vehicle response [ms], control accuracy [cm], ODD evidence, and commercial constraints.

Powertrain alternatives must be compared on duty-cycle efficiency, degradation, refuelling/charging downtime, mass, packaging, infrastructure, and total cost. Direct-hydrogen PEM fuel cells are not excluded by a 22% efficiency assumption; DOE lists approximately 60% electrical efficiency for direct-H2 PEM. MS-SOFC remains the selected base-case candidate for this concept, subject to full TCO, durability, start-up, emissions, and ODD validation.

Node operation requires 2,080.5kWh/day of net traction energy. At 60% efficiency the fuel-LHV requirement is 3,467.5kWh/day, equivalent to 104.1kg-H2/day. PV generation of 1,402,500kWh/year gives an annual energy ratio of approximately 1.85, but continuous operation still requires an hourly supply-demand, storage, conversion-loss, and weather model.

"Motion sickness free" refers to a condition where control latency ≤20ms, decision latency ≤100ms, positioning error ≤10cm, and vibration ≤0.3Hz are simultaneously satisfied. This condition determines user acceptance and directly enables Utilization = 0.75 (=18h/day ÷ 24h/day). HPC maintains control at an execution rate of 100Hz (=1 ÷ 0.01sec), ensuring continuous satisfaction of response constraints. The definition is retained as the design requirement; compliance must be demonstrated by test evidence.

Block	Variable	Value / status	Formula / validation	KPI link
KPI	Energy Efficiency	0.58 target	metered useful output ÷ purchased input	Energy
	Utilization	0.75 target	18÷24	Operation
	Load Factor	0.70 target	2.8÷4	Demand
Constraint	Operation Hours	24h/day target	operating window	Utilization
	Stability	0.95 target	1−0.05	Energy
	Throughput	40veh/h	1 vehicle ÷1.5min	Infra
Node	Net energy	2,080.5kWh/day	28.5×73	Energy
	Fuel input	3,467.5kWh/day	2,080.5÷0.60	Energy
	SOFC efficiency	0.60 target	net electricity ÷ fuel LHV	Energy
	H2 LHV	33.3kWh/kg	constant	Energy
	H2 required	104.1kg/day	3,467.5÷33.3	Supply
	Annual net energy	759,383kWh/yr	2,080.5×365	Energy
	PV capacity	1,500kW	1.5MW×1,000	Energy
	PV yield	1,100kWh/kW/yr	benchmark assumption	Energy
	PV loss	0.15	15%	Energy
	PV generation	1,402,500kWh/yr	1,500×1,100×0.85	Energy
	Annual PV ratio	1.85	1,402,500÷759,382.5	Annual only
	Continuous supply	Not yet demonstrated	hourly balance + storage + reserve	Utilization
	Supply stability	0.95 target	supplied ÷ required	Energy
	Node delay	≤50ms target	measured	Infra
	Dispatch interval	5sec	60÷12	Utilization
	Trips	73/day target	scenario target	Demand
	Fleet	13	ceil(73÷6)	Operation
	Load	4pax	2×2	Demand
	Energy/trip	28.5kWh	2,080.5÷73	Energy
	Reliability	0.99 target	successful cycles ÷ total	Stability
	Availability	0.99 target	available ÷ scheduled time	KPI
Vehicle	Control Sync	≤20ms	100ms÷5	Motion sickness free
	Decision Sync	≤100ms	100ms	Motion sickness free
	Position Error	≤10cm	1m÷10	Motion sickness free
	Platoon Stability	≤5%	0.05	Motion sickness free
	Control Cycle	100ms	1sec÷10	Stability
	Response	≤20ms	100ms÷5	Motion sickness free
	Jerk	≤0.5m/s²/s	1÷2	Motion sickness free
	Vibration	≤0.3Hz	1Hz÷3.33	Motion sickness free
	Sync Accuracy	0.99	1−0.01	Stability
Control	Prediction	1.00 target	successful prediction ratio	Operation
	Planning	1.00 target	successful plan ratio	Operation
	Execution	100Hz	1÷0.01sec	Control
	HPC latency	≤10ms target	measured	Stability
	Node sync	≤50ms target	measured	Stability
	Flow	40veh/h	60÷1.5	Infra
UX	Utilization Gain	0.10 target	0.75−0.65	KPI

This defines the conditions under which autonomous mobility may become operationally feasible. Passing one ratio does not substitute for validation of the remaining gates.

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Proposed Responsibility Structure for Autonomous Mobility System

This is a proposed RACI hypothesis for discussion, not a statement of current accountability of any specific company or consortium. Final responsibility must be confirmed by corporate governance, contracts, permits, product-liability allocation, operational rules, and the approved safety case.
Continuous operation depends on explicit ownership across orchestration, fleet operation, vehicle/system safety, node execution, energy supply, commercial settlement, and finance. A method such as model-based development is not an accountable legal entity; each task must be assigned to an organization or contracted operator.

Domain	Function / task	Accountable (A)	Responsible (R)	Support / consulted	Status
Demand	Generate demand	Mobility Platform	Mobility Platform	AD Platform / partners	Proposed
	Capture demand	Mobility Platform	Mobility Platform	MaaS Operator	Proposed
	Set pricing	Commercial owner	Commercial owner	Finance / regulator	Contract required
	Approve pricing	Vehicle OEM / business owner	Commercial owner	Legal / regulator	Approval required
	Optimize load	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Manage UX	Mobility Platform	Mobility Platform	AD Platform / users	Proposed
	Define service	Vehicle OEM / business owner	Mobility Platform	MaaS Operator / user	Proposed
Operation	Execute dispatch	MaaS Operator	MaaS Operator	Mobility / AD Platforms	Permit + SLA
	Operate fleet	MaaS Operator	MaaS Operator	OEM / AD Platform	Permit + SLA
	Maintain service	MaaS Operator	MaaS Operator	Node operator / Mobility Platform	SLA
	Manage utilization	MaaS Operator	Mobility Platform	AD Platform	Proposed
	Define hours	MaaS Operator	MaaS Operator	Regulator / node operator	Permit
	Control dispatch	MaaS Operator	AD control platform	Mobility Platform	Authority boundary
Vehicle	Define architecture	Vehicle OEM	Vehicle OEM	AD Platform / Tier-1	Product governance
	Supply platform	Vehicle OEM	Vehicle OEM	Tier-1 suppliers	Commercial contract
	Supply drivetrain	Vehicle OEM	Tier-1 supplier(s)	AD Platform	Commercial contract
	Package cabin	Vehicle OEM	Body integrator / supplier	Mobility Platform	Commercial contract
	Maintain vehicle	Fleet owner / MaaS Operator	Certified service provider	Vehicle OEM	SLA
	Realize control	Vehicle OEM	AD Platform + Tier-1	MaaS Operator	Safety case
	Maintain position	AD Platform / OEM	AD Platform	Mobility Platform	Safety case
Facility	Provide site	Node / Parking Operator	Node / Parking Operator	Vehicle OEM	Contract
	Operate facility	Node / Parking Operator	Node / Parking Operator	MaaS Operator	Permit + SLA
	Sell services	Commercial owner	Mobility Platform / provider	Node operator	Contract
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
	Manage circulation	Node operator	Node operator	Vehicle OEM	SLA
Service	Provide booking	Mobility Platform	Mobility Platform	AD Platform	Service governance
	Integrate service	Mobility Platform	Mobility Platform	Service providers	API contract
	Sell package	Commercial owner	Commercial owner	Finance	Contract
	Optimize stays	MaaS Operator	Mobility Platform	Service providers	Proposed
	Operate service	Service provider	Service provider	MaaS Operator	SLA
MaaS	Operate platform	MaaS Operator	AD + Mobility Platforms	Vehicle OEM	Authority boundary
	Use platform	MaaS Operator	MaaS Operator	AD Platform	SLA
External	Provide entry	External platform owner	External platform owner	Mobility Platform	API contract
	Send traffic	External platform owner	External platform owner	MaaS Operator	Commercial contract
	Receive traffic	MaaS Operator	Mobility Platform	External platform	Commercial contract
Energy	Define supply	Energy business owner	Energy integrator / supplier	AD Platform / node operator	Contract
	Produce hydrogen	Energy supplier	Energy supplier	Node operator	Permit + quality
	Distribute energy	Energy supplier	Energy integrator / supplier	Node operator	Contract
	Manage logistics	Energy Integrator	Energy Integrator	Energy supplier	Contract
	Operate energy	Node operator	Node operator	AD Platform	SLA
Software	Define scope	Vehicle OEM / product owner	AD Platform	Mobility Platform	Product governance
	Develop control	AD Platform	AD Platform	Vehicle OEM / Tier-1	Release governance
	Operate compute	AD Platform	Cloud / edge operator	MaaS Operator	SLA
	Maintain dispatch	MaaS Operator	AD + Mobility Platforms	Node operator	SLA
Finance	Collect mobility	Commercial owner	Finance / payment operator	Mobility Platform	Settlement contract
	Collect facility	Node operator	Finance / payment operator	Mobility Platform	Settlement contract
	Collect energy	Energy business owner	Finance / payment operator	Energy Integrator	Settlement contract
	Maintain projection	Business owner	Finance	Vehicle OEM	Governance
	Collect data	Data controller	Mobility Platform	AD Platform / operators	Privacy + data contract
	Collect facility data	Node operator	Mobility Platform	AD Platform	Privacy + data contract

Responsibility is complete only after the accountable legal entity, operational authority, safety authority, data controller, and revenue/cost owner are all confirmed.

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Pricing Structure and Billing Rules for Autonomous Mobility System

The amounts below remain illustrative scenario tariffs, not verified market prices. To prevent double counting, one base trip fare is selected per trip; add-ons apply only when the corresponding service is consumed; subscriptions offset eligible base charges; B2B platform fees are charged only when not already embedded in another contract; dividends/revenue shares are settlements below revenue, not customer prices.

Category	Payer	Payee	Charge	Billing rule	Illustrative formula / annualization	P&L mapping
User	User	Service provider	Facility Fee	Add-on	¥500/visit×200/day×365 = ¥36.500M/yr	MaaS service
			Premium Service	Add-on	¥1,500/booking×120/day×365 = ¥65.700M/yr	Service revenue
			Add-on	Add-on	¥600/use×200/day×365 = ¥43.800M/yr	Service revenue
		MaaS Operator	Service Fee	Base fare (exclusive)	¥1,000/trip×130/day×365 = ¥47.450M/yr	Trip revenue
			Trip Fee	Base fare (exclusive)	¥2,000/trip×100/day×365 = ¥73.000M/yr	Trip revenue
			Premium Trip	Base fare (exclusive)	¥2,500/trip×90/day×365 = ¥82.125M/yr	Trip revenue
			Subscription	Offsets eligible base fares	¥15,000/user×1,200 users = ¥18.000M/yr	Subscription revenue
			Night Trip	Base fare (exclusive)	¥2,700/trip×80/day×365 = ¥78.840M/yr	Trip revenue
		Parking Operator	Premium Facility	Add-on	¥800/visit×120/day×365 = ¥35.040M/yr	Facility revenue
			Loyalty Fee	Add-on / retention	¥200/visit×185/day×365 = ¥13.505M/yr	Service revenue
Facility	AD Platform	Parking Operator	Site Rent	B2B fixed fee	¥300,000/month×30×12 = ¥108.000M/yr	Node OPEX
			Operating Access	B2B access fee	¥700/visit×90/day×365 = ¥22.995M/yr	Node OPEX
	Parking Operator	Service Fee	B2B service	¥300/visit×200/day×365 = ¥21.900M/yr	Node revenue	
	Service Provider	Parking Operator	Upgrade Rent	B2B fixed fee	¥2,000,000/month×30×12 = ¥720.000M/yr	Node OPEX
Service	Service Provider	Mobility Platform	Booking Fee	Per booking	¥100/booking×150/day×365 = ¥5.475M/yr	Reservation fee
			CRM Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee
			API Fee	B2B usage fee	¥80/call×250/day×365 = ¥7.300M/yr	Platform fee
			Premium UX	B2B product fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee
		AD Platform	Optimization Fee	B2B product fee	¥50/call×300 calls/day×365 = ¥5.475M/yr	Platform fee
			Premium API	B2B usage fee	¥150/call×60/day×365 = ¥3.285M/yr	Platform fee
			Sync Fee	B2B product fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee
		Service Provider	Late Fee	Customer charge	¥1,200/use×70/day×365 = ¥30.660M/yr	Service revenue
Mobility	MaaS Operator	AD Platform	Control Fee	B2B product fee	¥100/dispatch×100/day×365 = ¥3.650M/yr	Control OS
			Routing Fee	B2B product fee	¥120/dispatch×110/day×365 = ¥4.818M/yr	Platform fee
		Mobility Platform	UX Fee	B2B product fee	¥50/trip×100/day×365 = ¥1.825M/yr	Platform fee
		Parking Operator	Advanced Control	B2B service fee	¥150/dispatch×90/day×365 = ¥4.928M/yr	Node service
	AD Platform	MaaS Operator	Incentive	Contra-revenue / settlement	¥200,000/month×30×12 = ¥72.000M/yr	Settlement; not gross revenue
	User	MaaS Operator	Surge Pricing	Base fare adjustment	¥2,800/trip×80/day×365 = ¥81.760M/yr	Trip revenue
			Off-peak Discount	Base fare adjustment	¥1,600/trip×70/day×365 = ¥40.880M/yr	Trip revenue
Platform	MaaS Operator	AD Platform	OS Fee	B2B fixed fee	¥80,000/month×30×12 = ¥28.800M/yr	Platform fee
			Monthly OS	B2B fixed fee	¥250,000/month×30×12 = ¥90.000M/yr	Platform fee
	Service Provider	AD Platform	Node Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service
			Data Fee	B2B usage fee	¥100,000/month×30×12 = ¥36.000M/yr	Platform fee
	Parking Operator	AD Platform	Facility Fee	B2B fixed fee	¥200,000/month×30×12 = ¥72.000M/yr	Node service
			Analytics Fee	B2B fixed fee	¥120,000/month×30×12 = ¥43.200M/yr	Platform fee
			Control OS	B2B product fee	¥220/trip×110/day×365 = ¥8.833M/yr	Control OS
			Booking Link	B2B usage fee	¥180,000/month×30×12 = ¥64.800M/yr	Reservation fee
External	External Platform	AD Platform	API Fee	B2B usage fee	¥120/use×5,000/day×365 = ¥219.000M/yr	Platform fee
			IoT Fee	B2B usage fee	¥90/booking×200/day×365 = ¥6.570M/yr	Platform fee
			Tx Fee	B2B usage fee	¥200/booking×100/day×365 = ¥7.300M/yr	Platform fee
			Premium API	B2B usage fee	¥150/call×6,000/day×365 = ¥328.500M/yr	Platform fee
			Premium Tx	B2B usage fee	¥250/booking×180/day×365 = ¥16.425M/yr	Platform fee
			Premium Referral	Revenue share / settlement	15% of eligible booking revenue	Settlement; not gross revenue
			Sync Fee	B2B usage fee	¥12/call×5,500/day×365 = ¥24.090M/yr	Platform fee
Energy	AD Platform	Finance Co.	Energy Fee	B2B service	20kWh/trip×100/day×¥30×365 = ¥21.900M/yr	Energy revenue
			Peak Energy	B2B service	25kWh/trip×90/day×¥45×365 = ¥36.956M/yr	Energy revenue
			Smart Energy	B2B service	¥70/use×120/day×365 = ¥3.066M/yr	Energy revenue
	Service Provider	AD Platform	Energy Opt	B2B product fee	¥50/use×150/day×365 = ¥2.738M/yr	Platform fee
Finance	MaaS Operator	Finance Co.	Base Energy	B2B settlement	18kWh/trip×110/day×¥28×365 = ¥20.236M/yr	Energy OPEX
			Lease	B2B lease	¥3,000,000/year×365-day basis = contract specific	Lease / CAPEX
			Dividend	Settlement only	profit distribution after tax	Not revenue
			Node Lease	B2B lease	¥4,500,000/year per node	Lease / CAPEX
			Node Lease	B2B lease	¥2,500,000/year per node	Lease / CAPEX
	Service Provider	Finance Co.	Lease	B2B lease	¥2,000,000/year×365-day basis = contract specific	Lease / CAPEX
			Dividend	Settlement only	profit distribution after tax	Not revenue
			Service Lease	B2B lease	¥2,500,000/year per asset	Lease / CAPEX

Only lines explicitly mapped to the P&L may be included. Settlement lines and mutually exclusive base fares must not be added again.

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Autonomous Mobility Business Control Structure

System conditions define the feasibility of control and set operational boundaries. Values below are phase targets or scenario assumptions; they are not externally verified outcomes. Minimum gates on P1 and phase targets here are different by design and must be labelled separately.
Revenue is calculated with unit-consistent formulas. When fare is per passenger, annual revenue = Trips/day × Seats × Load factor × Fare/passenger × 365. Break-even trips/day cannot be calculated without a defined annual controllable-cost allocation.

Layer	Control item	Decision logic	2030	2035	P&L impact
Demand	Demand generation	Requests ÷ potential demand	≥0.50	≥0.70	Trips
	Conversion	Bookings ÷ requests	≥0.70	≥0.85	Trips
	Matching	Matched ÷ requests	≥0.85	≥0.95	Trips
	Trip realization	Executed ÷ matched	≥0.80	≥0.95	Trips
	Stabilization	Matched trips + planned demand	≥0.80	≥0.95	Trips
	Time synchronization	Calendar-based demand	≥0.60	≥0.85	Trips
	Service capture	Reservation-based trips ÷ total	≥0.50	≥0.75	Trips
	Demand-cap control	Demand ≤ system capacity	100%	100%	Trips cap
Operation	Trips	Executed trips/day	73	140	Trips
	Utilization	Operating time ÷ 24h	≥0.75	≥0.85	Trips
	Dispatch success	Successful dispatch ÷ requests	≥0.95	≥0.99	Trips
	Supply-body cycle	Active body ÷ total cycle	≥0.80	≥0.95	Utilization
	Waiting time	Passenger request-to-pickup	≤360sec	≤180sec	Trips
	Turnaround	Idle + service time	≤25min	≤10min	Utilization
	Queue control	Total wait ÷ requests	≤300sec	≤120sec	Conversion
	Optimization	Served ÷ queued	≥0.90	≥0.98	Trips
	Scheduled continuity	Continuous-operation ratio	≥0.80	≥0.95	Utilization
	Operation cap	Operation ≤24h	Fixed	Fixed	Cap
Vehicle	Latency control	Control delay	≤100ms	≤50ms	Dispatch
	Stability	Deviation ÷ target	≤15%	≤5%	Safety
	Intervention	Interventions ÷ km	≤0.010/km	≤0.0001/km	Utilization
	Motion response	Control latency / planning / ride targets	Design gate	Design gate	Utilization
	Response accuracy	Correct ÷ total control	≥0.90	≥0.98	Dispatch
	Platoon composition	Vehicles per formation	≥1.5	≥3.0	Load
	Ride continuity	Continuous ride + trips	≥0.85	≥0.90	Utilization
	Thermal constraint	Temperature limit	Fixed	Improved	Operation
	Vehicle availability	Available fleet ÷ total	≥0.80	≥0.95	Trips
Energy	End-to-end efficiency	Useful output ÷ purchased input	0.58 target	0.65 target	Cost
	Energy cost/trip	Cost ÷ trips	≤¥500	≤¥350	Cost
	Supply stability	Continuous supplied ÷ required	≥0.95	≥0.99	Utilization
	H2-equivalent requirement	Net energy ÷ efficiency ÷33.3	104.1kg/day	171.2kg/day	Trips cap
	PV role	Annual energy contribution	Annual ratio only	Annual ratio only	Cost
	Storage capacity	Hourly storage / reserve	Required	Required	Trips cap
	Distribution	Loss + supply	≤10%	≤5%	Cost
	Node allocation	Allocated ÷ demand	≥0.90	≥0.98	Trips
	Energy scheduling	Scheduled + required	≥0.85	≥0.95	Trips
System	Trip generation	Demand×Matching×Dispatch×Utilization	73/day	140/day	Revenue
	Load optimization	Passengers ÷ capacity	0.70	0.80	Revenue
	Fare	JPY per passenger	¥800	¥900	Revenue
	Throughput	Trips ÷ hour	≥1.5	≥3.0	Revenue
	Density	Trips ÷ area	Low	High	Revenue
	Scaling factor	Trips 2035 ÷ 2030	1.00×	1.92×	Revenue
	Energy linkage	28.5kWh × trips	2,080.5kWh	3,990kWh	Cost
	Hydrogen linkage	Energy ÷ efficiency ÷33.3	104.1kg	171.2kg	Cost
	Fleet sizing	ceil(trips ÷6)	13	24	Trips cap
	Congestion	Load ÷ capacity	Medium	High	Limit
	Node capacity	Throughput ≤ capacity	Limited	Expanded	Cap
Financial	Annual passenger revenue	Trips×Seats×Load×Fare×365	¥59.685M	¥183.960M	Profit
	Revenue per vehicle trip	Seats×Load×Fare	¥2,240	¥3,600	Revenue
	Contribution	Allocated revenue − allocated cost	See rates	See rates	IRR
	Break-even trips/day	Annual controllable cost ÷ (Seats×Load×Fare×365)	TBD	TBD	Profit
	Cost control	Energy + operation + vehicle + infrastructure	High	Reduced target	Cost
	Price constraint	Applicable fare / contract rules	Validate	Validate	Revenue
	IRR hurdle	Return target	≥10%	≥12%	Investment

This defines unit-consistent control logic connecting operational conditions to trips, revenue, cost, and investment gates.

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Autonomous Mobility Business Profit & Loss Structure (2030 / 2035)

This is a scenario P&L, not a forecast. Each monetary line is recalculated from the displayed formula. Non-monetary operating drivers are separated from revenue. The prior ¥36B node-sale line is excluded from the base case because asset-transfer price, delivery cost, and COGS were not defined; recurring node-operation service remains included.

Layer	Category	Item	2030 formula	2030	2035 formula	2035
Revenue	Energy	Elec Internal	4.0M×20%×50%×1,200×¥15	¥7.200B	4.0M×30%×50%×1,500×¥15	¥13.500B
		Elec External	4.0M×25%×50%×1,200×¥15	¥9.000B	4.0M×35%×50%×1,500×¥15	¥15.750B
		H2 Internal	4.0M×80%×1%×67×¥333	¥0.714B	4.0M×70%×20%×67×¥333	¥12.494B
		H2 External	4.0M×75%×1%×67×¥333	¥0.669B	4.0M×65%×20%×67×¥333	¥11.602B
		Titanium	80,000×0.3×¥72,000	¥1.728B	same	¥1.728B
		HPC Compute Service	600 GPU×8,760×35%×¥700	¥1.288B	5,400 GPU×8,760×55%×¥600	¥15.610B
Driver		24H Utilization	18÷24	0.75	20.4÷24	0.85
		Load Factor	2.8÷4	0.70	4.0÷5	0.80
		Swap-body Util.	Active÷Total	0.80	Dynamic composition	0.95
		Night Operation	Night revenue÷Total	0.35	Continuous operation	0.60
		Auto Loading Rev.	Loading fee rev.÷Total	0.22	Integrated logistics	0.35
Revenue	MaaS	Night Passenger	25×24×¥15,000×365×60%	¥1.971B	500×24×¥15,000×365×60%	¥39.420B
		Night Cargo	25×24×¥5,000×365×40%	¥0.438B	500×24×¥5,000×365×40%	¥8.760B
		Mobile Storage	400×¥40,000×12	¥0.192B	5,000×¥40,000×12	¥2.400B
	Platform	Auto Loading	269,000×¥22,300	¥5.999B	538,000×¥22,300	¥11.997B
		Platform Fee	Dispatch/API fee	¥0.800B	Integrated OS fee	¥6.500B
		Reservation Fee	Booking linkage	¥0.350B	Integrated booking	¥2.200B
		Node Service	Node operation fee	¥0.420B	Node ecosystem service	¥4.800B
Total		Total Revenue	sum monetary revenue	¥30.769B	sum monetary revenue	¥146.761B
OPEX	Energy	H2 Internal COGS	4.0M×80%×1%×67×¥150	▲¥0.322B	4.0M×70%×20%×67×¥150	▲¥5.628B
		H2 External COGS	4.0M×75%×1%×67×¥150	▲¥0.302B	4.0M×65%×20%×67×¥150	▲¥5.226B
		Elec Internal	4.0M×20%×50%×1,200×¥6.7	▲¥3.216B	4.0M×30%×50%×1,500×¥6.7	▲¥6.030B
		Elec External	4.0M×25%×50%×1,200×¥6.7	▲¥4.020B	4.0M×35%×50%×1,500×¥6.7	▲¥7.035B
		Titanium	80,000×0.3×¥6,000	▲¥0.144B	same	▲¥0.144B
		Material Cycle	scenario cost	▲¥0.300B	onboard circulation	▲¥1.800B
		Node Optimization	distributed control	▲¥0.120B	integrated mesh	▲¥0.950B
		Energy Sync	In/Out sync	▲¥0.080B	real-time optimization	▲¥0.600B
	MaaS	MaaS Ops	25×¥81.0M	▲¥2.025B	500×¥45.6M	▲¥22.800B
		Dispatch Ops	fleet dispatch	▲¥0.550B	integrated dispatch	▲¥4.200B
		Remote Ops	remote support	▲¥0.300B	integrated operation	▲¥2.100B
	Platform	Platform Ops	¥1.0B/year	▲¥1.000B	¥6.0B/year	▲¥6.000B
		Cloud Ops	base + power + service	▲¥0.731B	base + power + service	▲¥6.680B
		AI Ops	AI orchestration	▲¥0.220B	integrated AI control	▲¥2.400B
Total		Total OPEX	sum	▲¥13.329B	sum	▲¥71.593B
SG&A	Energy	Energy SG&A	energy product revenue×10%	▲¥1.931B	energy product revenue×10%	▲¥5.507B
		Maintenance	scenario	▲¥0.250B	integrated maintenance	▲¥1.800B
	MaaS	MaaS SG&A	MaaS revenue×15%	▲¥0.390B	MaaS revenue×15%	▲¥7.587B
		Customer Ops	user support	▲¥0.180B	integrated support	▲¥1.500B
	Platform	Platform SG&A	SW revenue×20%	▲¥1.514B	SW revenue×20%	▲¥5.099B
		Partner Ops	alliance operation	▲¥0.120B	integrated alliance	▲¥1.100B
Total		Total SG&A	sum	▲¥4.385B	sum	▲¥22.594B
Profit	Total	EBITDA	Revenue−OPEX−SG&A	¥13.054B	Revenue−OPEX−SG&A	¥52.574B
Dep.	Energy/MaaS/Platform	Node	20.000B÷15y	▲¥1.333B	86.400B÷15y	▲¥5.760B
		GTFC	18.450B÷15y	▲¥1.230B	22.600B÷15y	▲¥1.507B
		Mill	0.600B÷10y	▲¥0.060B	4.800B÷10y	▲¥0.480B
		Battery	5.000B÷10y	▲¥0.500B	6.400B÷10y	▲¥0.640B
		Magnetic	3.000B÷10y	▲¥0.300B	3.800B÷10y	▲¥0.380B
		HTS	4.000B÷10y	▲¥0.400B	5.100B÷10y	▲¥0.510B
		Water	2.000B÷10y	▲¥0.200B	2.600B÷10y	▲¥0.260B
		HPC	5.000B÷15y	▲¥0.333B	45.000B÷15y	▲¥3.000B
		Energy Device	Included in Mill/Batt/HTS	¥0.000B	Included in Mill/Batt/HTS	¥0.000B
		Vehicle Dev	50.000B÷5y	▲¥10.000B	63.900B÷5y	▲¥12.780B
		Prototype	10.000B÷5y	▲¥2.000B	10.000B÷5y	▲¥2.000B
		Swap-body Dev	Included in Vehicle Dev	¥0.000B	Included in Vehicle Dev	¥0.000B
		Platform Dev	40.000B÷5y	▲¥8.000B	59.400B÷5y	▲¥11.880B
		AI Platform	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
		Control OS	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
		Infra Sync	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
		Revenue Logic	Included in Platform Dev	¥0.000B	Included in Platform Dev	¥0.000B
Total		Total Depreciation	sum	▲¥24.356B	sum	▲¥39.197B
Profit	Total	Operating Profit	EBITDA−Dep.	▲¥11.302B	EBITDA−Dep.	¥13.377B
		Contribution before depreciation	Revenue−OPEX−SG&A	¥13.054B	Revenue−OPEX−SG&A	¥52.574B
		Operating Margin	Operating Profit÷Revenue	-36.7%	Operating Profit÷Revenue	9.1%
		Project Payback	see cash-flow table	not achieved	see cash-flow table	discounted not achieved by 2035
		Project IRR	reconciled cash flow	-12.13%	12% hurdle	not met

Assumptions retained unless corrected above: city 1/5; MaaS vehicles 25/500; nodes 4/36; passenger cars 4.0M/year; trucks 80,000/year; electricity sales ¥15/kWh and cost ¥6.7/kWh; H2 sales ¥333/kg and cost ¥150/kg; titanium exchange 80,000×0.3, sales ¥72,000 and cost ¥6,000. These are scenario assumptions requiring primary-source validation.

Depreciation is rebased to the ¥310.000B total investment (nodes 36×¥2.4B; HPC unchanged). Energy Device, Swap body, AI, Control OS, Infra Sync and Revenue Logic are included in their parent asset rows. Cash flow, FCF, IRR and NPV are unchanged.

Satofumi Goto

2030 Vehicle Platform

A 12m autonomous MaaS skateboard uses an MS-SOFC scenario with city gas 70% and mill-derived hydrogen 30%. Hydrogen is supplied from stationary/node-side milling; batteries support transient power. Component ranges sum to approximately 9.8–15.1t before the stated 10–15% integration reserve; the upper case approaches 17.3t. Therefore, payload is not claimed until body, fluids, passengers, wiring/piping, axle loads, and certified curb mass are included.

The 10–15kg LH2 buffer requires approximately 340L after the 1.6 packaging factor. The envelope is corrected to 450×550×1,400mm (346.5L external envelope). For 240kW electrical output at 60% efficiency, fuel input is 400kW and nominal rejected heat is approximately 160kW, not 96kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H2/Power		12m flat skateboard	-	upper container unused	-	-	overall
Total energy demand	1	1.2kWh/km	1.2×1,000	1,200kWh	-	-	overall
Required fuel energy	1	MS-SOFC 60%	1,200÷0.60	2,000kWh-LHV	-	-	overall
Hydrogen ratio	1	H2 30% blend	2,000×0.30	600kWh-H2	-	-	overall
Required H2	1	33.3kWh/kg	600÷33.3	18.0kg-H2	-	-	overall
City-gas ratio	1	70%	2,000×0.70	1,400kWh-gas	-	-	overall
LH2 buffer	1	10–15kg	reserve	10–15kg-H2	450/550/1400	150–250kg	left
LH2 installed volume	1	insulation×1.6	15÷70.8×1.6	≈340L	450/550/1400	included	left
City-gas tank	2	blended-fuel support	-	operation reserve	300/400/1600	300–500kg	left
Water tank	2	cooling/process water	-	≈180L	250/400/1200	180kg	left
Process-water control	2	steam-to-carbon / condensate	validated setpoint	not NOx claim	180/250/400	80–120kg	front
O2 membrane	2	optional O2 management	test gate	efficiency TBD	180/250/450	80–120kg	front
CO2 membrane	2	reformate cleanup	70–90% target	CO2 removal target	220/250/500	100–150kg	rear
Steam membrane	2	H2O separation	test gate	reforming support	180/250/450	80–120kg	rear
Reformer	2	steam reforming	fuel-path model	H2+CO reformate	260/350/700	250–350kg	front
Gas/liquid separator	2	50L/min	50×2	100L/min	220/250/450	120–180kg	F/R
H2 purifier	2	50L/min	50×2	100L/min	220/250/500	120–180kg	F/R
Vaporizer	2	20kg/h	20×2	40kg/h	240/300/600	120–180kg	left
LH2 pump	2	50L/min	50×2	100L/min	180/220/350	60–100kg	left
Pressure regulator	4	high→low P	-	4 systems	120/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	120/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1400	20–40kg	upper
Safety sensors	1	H2/P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		MS-SOFC power control	-	battery transient	-	-	center
MS-SOFC modules	6	40kW/unit	40×6	240kW	320/1000/1400	1,200–1,800kg	center
MS-SOFC efficiency	1	electrical efficiency	-	≈60% target	-	-	center
Inverter	2	120kW	120×2	240kW	180/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	160/350/500	120–180kg	upper center
Solid-state battery	1	400kWh	-	400kWh	260/1200/2200	2,500–3,500kg	center
HV J-BOX	1	HV distribution	-	1 system	180/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	120/250/300	40–80kg	center
Radiator	2	rejected heat	240×(1/0.60−1)	≈160kW	180/500/1400	180–300kg	front
Cooling pump	2	50L/min	50×2	100L/min	160/180/250	40–80kg	overall
Waste-heat recovery	1	Rankine/thermal	validated recovery	aux. power TBD	260/600/900	250–450kg	rear
Insulation monitoring	1	HV safety	-	1 system	80/200/300	<20kg	center
Power protection	1	fuse/breaker	-	integrated	80/200/300	<20kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	prediction/planning	latency≤100ms	control target	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 Drivetrain		electric drive	-	AD ride stability	-	-	corners
IWM	4	80kW/wheel	80×4	320kW	320/500/500	1,000–1,400kg	corners
Regen brake	4	regenerative	-	4 wheels	400/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	400/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	260/320/420	250–350kg	corners
Air suspension+CFRP	4	vibration suppression	-	4 wheels	260/350/450	250–350kg	corners
CTIS tire	4	pressure control	-	4 wheels	220/500/500	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	550–600mm target	-	-	overall
Crossmember	6	structural beam	-	6 beams	160/1600/160	250–400kg	lateral
MH main chassis	2	H2 storage+frame	-	≈10kg-H2	400/350/5500	400–800kg	longitudinal
LH2 subframe	2	cooling+support	-	2 systems	260/450/2200	300–500kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	50–100/car	400–700kg	lower/center
Container lock	4	swap fixation	-	4 points	180/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	180/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	gas/LH2/water	150/500/900	<120kg	L/R
Boarding assist	1	kneel+2-step	−100mm	access assist	220/300/800	120–200kg	right
Payload margin	1	25t GVW−verified curb mass	full integration required	TBD; prior 10–13t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

Satofumi Goto

2035 Vehicle Platform

A 12m platform targets a 380–400mm low floor. The platform-weight and 13–16t payload claims are removed pending full integrated mass, body, fluids, passengers, wiring/piping, axle-load, and certification data. The energy balance now separates the 400kWh air-battery contribution from PCFC energy demand.

PCFC-only hydrogen demand is 48.1kg/day (=1,200÷33.3÷0.75). With a 400kWh air-battery contribution, net PCFC energy is 800kWh and net H2 demand is 32.0kg/day. Six mills at 0.865kg/h for 12.5h produce 64.9kg/day, leaving a 32.9kg/day gross surplus before process losses. The 10–15kg LH2 buffer still requires about 340L; the envelope is corrected to 350×850×1,200mm (357L). Rejected heat at 240kW and 75% efficiency is approximately 80kW.

Component	Qty	Performance	Formula	Corrected result	H/W/D mm	Weight	Layout
1 Energy/H2/Power		system demand	1.2×1,000	1,200kWh	-	-	overall
Gross H2 before air battery	1	PCFC 75%	1,200÷(33.3×0.75)	48.1kg-H2	-	-	overall
Air-battery energy	4	100kWh/unit	100×4	400kWh	120/350/1200	≈849kg	right
H2 reduction equivalent	1	25kWh/kg-H2	400÷25	16.0kg-H2 eq.	-	-	right
Net PCFC energy	1	1,200−400	-	800kWh	-	-	overall
Net H2 required	1	PCFC 75%	800÷(33.3×0.75)	32.0kg-H2	-	-	overall
Hydrogen generation	6	0.865kg/h	0.865×6×12.5	64.9kg-H2	240/450/650	720–1,080kg	F/R
Gross surplus	1	before process losses	64.9−32.0	32.9kg-H2	-	-	overall
LH2 buffer	1	10–15kg reserve	-	10–15kg-H2	350/850/1200	100–150kg	left
LH2 installed volume	1	insulation×1.6	15÷70.8×1.6	≈340L	350/850/1200	included	left
Water tank	6	9kg/kg-H2 gross	64.9×9	584L	200/400/1200	584kg	left
Slurry tank	6	water×1.4	584×1.4	818L	230/450/1300	980–1,227kg	left
Titanium cartridge	6	3kg/0.865kg-H2	64.9÷0.865×3	≈225kg	160/300/800	225kg	L/R
PCFC modules	6	40kW/unit	40×6	240kW	180/900/1200	720–1,080kg	upper center
PCFC efficiency	1	electrical target	-	75% target	-	-	center
AMR liquefier	2	surplus-H2 process	10kg/day×2	process target	240/450/800	250–400kg	F/R
H2 purifier	3	50L/min	50×3	150L/min	180/250/500	120–180kg	F/R
Gas/liquid separator	3	50L/min	50×3	150L/min	200/250/450	120–180kg	F/R
Process-water management	2	condensate / humidity balance	validated setpoint	closed-loop target	150/250/400	80–120kg	front
O2 management membrane	2	optional cathode control	test gate	efficiency TBD	150/250/450	80–120kg	front
CO2 membrane	0	direct-H2 baseline	not required	excluded	-	0kg	rear
Steam separator	2	H2O separation	water balance	condensate recovery	150/250/450	80–120kg	rear
Reformer	0	direct-H2 baseline	not required	excluded	-	0kg	front
Anode loop	1	fuel recirculation	20% target	efficiency gain target	180/250/500	80–120kg	center
LH2 pump	2	50L/min	50×2	100L/min	150/220/350	60–100kg	left
Vaporizer	2	20kg/h	20×2	40kg/h	180/300/600	100–160kg	left
Pressure regulator	4	high→low P	-	4 systems	100/200/300	40–60kg	overall
Shutoff valve	6	safety shutoff	-	6 systems	100/180/250	40–60kg	overall
Vent line	2	safety vent	-	2 systems	100/100/1200	20–40kg	upper
Safety sensors	1	H2/P/T monitor	-	integrated	80/200/300	<30kg	overall
2 Electric/Thermal		PCFC power control	-	battery transient	-	-	center
Inverter	2	120kW	120×2	240kW	150/420/650	180–260kg	upper center
DC/DC	2	100kW	100×2	200kW	130/350/500	120–180kg	upper center
Response battery	1	80kWh	-	80kWh	120/900/1600	350–500kg	upper center
HV J-BOX	1	HV distribution	-	1 system	150/350/500	40–80kg	center
EMS/VCU	1	energy+vehicle ctrl	-	1 system	100/250/300	40–80kg	center
Radiator	2	rejected heat	240×(1/0.75−1)	≈80kW	120/400/1200	120–200kg	front
Cooling pump	2	50L/min	50×2	100L/min	120/180/250	40–80kg	overall
HTS cooling line	1	LH2 cooling	-	1 system	60/60/12000	80–120kg	frame
HTC/HTS cable	1	low-loss power	-	1 system	50/50/12000	60–120kg	center→wheel
Waste-heat recovery	1	TE/TPV/Rankine	validated recovery	aux. power TBD	220/500/800	200–350kg	rear
Waste-heat utilization	1	efficiency gain	-	target only	-	-	rear
Power protection	1	insulation/fuse	-	integrated	80/200/300	<30kg	center
3 AD Control		autonomous execution	-	AD operation	-	-	upper
AD computer	1	planning/control compute	validated ODD load	performance TBD by evidence	120/400/500	40–80kg	upper
Sensor suite	1	LiDAR/camera/radar	-	360° sensing	120/500/800	40–80kg	upper
Fusion/localization	1	map+IMU+GNSS	-	position control	80/300/400	<30kg	upper
ODD/DDT monitor	1	state-dependent safety gate	condition class	HOLD / restrict / MRM / STOP	80/250/300	<20kg	center
Dispatch/routing ECU	1	trip execution	-	fleet control	100/300/400	20–40kg	upper
ETA/latency control	1	synchronization	latency≤50ms	AD stability target	80/250/300	<20kg	upper
4 HTS Drivetrain		SC drive+ride ctrl	-	no-sick UX	-	-	corners
LH2 SC-IWM	4	80kW/wheel	80×4	320kW	260/450/450	800–1,200kg	corners
SC regen brake	4	regen braking	-	4 wheels	300/400/400	120–180kg	corners
Air brake	4	braking	-	4 wheels	300/400/400	120–180kg	corners
Knuckle+EPS	4	steer support	-	4 wheels	220/300/400	200–300kg	corners
Air sus.+CFRP	4	vibration cut	-	4 wheels	220/300/400	200–300kg	corners
CTIS/CFRP wheel	4	pressure+light wheel	-	4 wheels	200/450/450	included	corners
Ride comfort target	1	no motion sickness	lateral accel control	UX stabilization	-	-	cabin
5 Structure		flat payload frame	-	380–400mm target	-	-	overall
HTS crossmember	6	beam+cooling	-	6 beams	120/1600/120	200–300kg	lateral
MH main chassis	2	H2 storage+frame	-	≈20kg-H2	300/300/5500	300–600kg	longitudinal
LH2/HTS subframe	2	cooling+support	-	2 systems	200/400/2000	200–400kg	lower
Protection/firewall	1	impact+fire safety	-	integrated	30–80/vehicle	300–600kg	lower/center
Container lock	4	swap fixation	-	4 points	150/300/300	80–120kg	upper
Swap positioning	4	±mm alignment	-	4 points	150/200/200	80–120kg	upper
Service interface	1	refill/drain/inspect	-	water/Ti/slurry	120/400/800	<100kg	L/R
Boarding assist	1	kneel+2-step	−100mm	access assist	200/300/800	120–200kg	right
Payload margin	1	25t GVW−verified curb mass	full integration required	TBD; prior 13–16t claim removed	-	-	overall
Service margin	1	maintenance reserve	total×10–15%	reserve	-	included	overall

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IRR and Cash Flow Structure

This page adds externally sold distributed-HPC capacity to the reconciled base cash flow. The node plan is 4/8/12/20/28/36; existing HPC CAPEX of ¥5B/¥45B implies ¥8.33M per installed unit and 150 H100-equivalent accelerators/node. Paid external utilization rises from 35% to 55% and blended price declines from ¥700 to ¥600 per accelerator-hour. Internal mobility compute is excluded from revenue. Incremental HPC OPEX includes 1.2kW per accelerator, PUE 1.15, ¥15/kWh electricity and service cost equal to 15% of HPC revenue. The case remains pre-tax and unlevered; working capital and cash taxes are zero.
IRR is the discount rate that sets NPV to zero. Project payback is the first year in which cumulative nominal FCF becomes non-negative. Discounted payback is assessed separately. Node payback cannot be calculated because node-level revenue, OPEX, CAPEX, tax, and renewal cash flow are not separately specified.

Investment and Depreciation Structure

Investment / Instrument	Timing	Dep.	Purpose
Initial Development Investment (2025–2030)			
Platform / AI / Urban OS	2025–2030	5y	Core control platform
MaaS Vehicle Prototype	2025–2028	5y	Vehicle feasibility
HTS / Battery / Mill	2026–2030	10y	Energy feasibility
Initial Nodes (4)	2027–2030	15y	Operational deployment
HPC / GTFC	2027–2030	15y	Control infrastructure
Expansion Investment (2030–2035)			
Node expansion (32; 36 total)	2030–2035	15y	City-scale deployment
Fleet Expansion (500)	2030–2035	5y	Utilization scaling
Distributed Energy Infrastructure	2030–2035	10–15y	Hydrogen/electricity scaling
AI / Dispatch Scaling	2030–2035	5y	Real-time optimization
Node Synchronization Infrastructure	2032–2035	15y	Distributed control

Displayed P&L Reference

Year	Revenue	EBITDA	Operating Profit	EBITDA Margin	Operating Margin
2030	¥30.769B	¥13.054B	▲¥11.302B	42.4%	-36.7%
2035	¥146.761B	¥52.574B	¥13.377B	35.8%	9.1%

Annual Cash Flow and Discounted Value

Cash-flow formulas: FCF_t = EBITDA_t − CAPEX_t − ΔWC_t − Cash Taxes_t; DCF_t = FCF_t/(1+r)^t; IRR solves Σ FCF_t/(1+IRR)^t = 0.

Year	Revenue	EBITDA	CAPEX	FCF	Discount factor 8%	Discounted FCF	Cumulative nominal FCF	Cumulative discounted FCF
2025	¥0.000B	▲¥5.000B	▲¥35.000B	▲¥40.000B	1.000	▲¥40.000B	▲¥40.000B	▲¥40.000B
2026	¥0.000B	▲¥6.000B	▲¥40.000B	▲¥46.000B	0.926	▲¥42.593B	▲¥86.000B	▲¥82.593B
2027	¥2.000B	▲¥5.000B	▲¥45.000B	▲¥50.000B	0.857	▲¥42.867B	▲¥136.000B	▲¥125.460B
2028	¥8.000B	▲¥2.000B	▲¥40.000B	▲¥42.000B	0.794	▲¥33.341B	▲¥178.000B	▲¥158.800B
2029	¥18.000B	¥3.000B	▲¥35.000B	▲¥32.000B	0.735	▲¥23.521B	▲¥210.000B	▲¥182.321B
2030	¥30.769B	¥13.054B	▲¥28.000B	▲¥14.946B	0.681	▲¥10.172B	▲¥224.946B	▲¥192.493B
2031	¥52.653B	¥19.893B	▲¥22.000B	▲¥2.107B	0.630	▲¥1.328B	▲¥227.053B	▲¥193.821B
2032	¥74.761B	¥26.910B	▲¥20.000B	¥6.910B	0.583	¥4.032B	▲¥220.143B	▲¥189.789B
2033	¥98.430B	¥35.199B	▲¥18.000B	¥17.199B	0.540	¥9.292B	▲¥202.944B	▲¥180.497B
2034	¥122.583B	¥43.880B	▲¥15.000B	¥28.880B	0.500	¥14.447B	▲¥174.064B	▲¥166.050B
2035	¥146.761B	¥52.574B	▲¥12.000B	¥40.574B	0.463	¥18.794B	▲¥133.489B	▲¥147.256B

Financial Results and Return Assessment

Financial result	Value	Interpretation	Decision meaning
Total CAPEX 2025–2035	¥310.000B	Sum of annual CAPEX	Pre-tax / unlevered model
Project IRR	-12.13%	Below 12% hurdle	Return hurdle not met
NPV @ 8%	▲¥147.256B	Discounted value	8% return-hurdle gap
NPV @ 12%	▲¥147.160B	Hurdle-rate value gap	Value uplift required
Nominal project payback	2039 (reference)	2035 FCF held flat	No renewal CAPEX/tax/WC
Discounted payback	Not achieved	Cumulative DCF remains negative	Separate scenarios required
Node payback	Not calculated	Node-level cash flow absent	Add node economics
2035 EBITDA Margin	35.8%	Displayed P&L	Cash generation before D&A
2035 Operating Margin	9.1%	Displayed P&L	Positive; IRR below hurdle

Sensitivity to Cash Flow Improvement

Sensitivity case	Explicit cash-flow change	IRR	NPV @ 8%	NPV @ 12%	Interpretation
No HPC external sales	HPC revenue excluded	-17.76%	▲¥165.852B	▲¥160.830B	Baseline comparison
HPC conservative	100 GPU/node; 40%×¥400/h	-15.71%	▲¥159.767B	▲¥156.351B	Not sufficient
Displayed HPC base	150 GPU/node; 55%×¥600/h	-12.13%	▲¥147.256B	▲¥147.160B	Improves; no payback
HPC high	200 GPU/node; 70%×¥900/h	-6.07%	▲¥118.914B	▲¥126.359B	Still insufficient
NPV@8% break-even	+¥43.337B/y in 2030–2035	8.00%	¥0.000B	▲¥33.927B	Beyond HPC alone

Nominal Cumulative FCF and Payback Extension

Period	Year / cumulative FCF	Year / cumulative FCF	Year / cumulative FCF
2025–2027	2025 / ▲¥40.000B	2026 / ▲¥86.000B	2027 / ▲¥136.000B
2028–2030	2028 / ▲¥178.000B	2029 / ▲¥210.000B	2030 / ▲¥224.946B
2031–2033	2031 / ▲¥227.053B	2032 / ▲¥220.143B	2033 / ▲¥202.944B
2034–2036	2034 / ▲¥174.064B	2035 / ▲¥133.489B	2036 / ▲¥92.915B
2037–2039	2037 / ▲¥52.341B	2038 / ▲¥11.767B	2039 / ¥28.807B PAYBACK
Extension basis	2036–2039	FCF ¥40.574B/year	No renewal CAPEX/tax/WC

Cash-Flow Completeness and Production-Case Inputs

Completeness item	Current treatment	Required production-case addition	Financial effect
Working capital	0	Receivables/payables/inventory timing by year	FCF / payback
Cash tax	0	Loss carry-forward, tax rate and timing	FCF / NPV
Financing	Excluded	Debt/equity, interest, fees and repayment	Equity IRR / DSCR
Residual / salvage	Excluded	Terminal asset value by asset class	Terminal FCF / NPV
Maintenance / renewal	Not separately shown	Overhaul, replacement and lifecycle cadence	OPEX / CAPEX
Node/HPC cash flow	Not separately shown	Revenue/OPEX/CAPEX/tax/LH2 cold energy by node	Node/HPC payback / ROIC

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Financial Interpretation & KPI Linkage

The displayed case adds a separately priced distributed-HPC service without counting internal mobility compute as revenue. 2035 HPC revenue is ¥15.610B and incremental HPC EBITDA is ¥12.730B. Annual project FCF is positive from 2032, but cumulative nominal FCF remains ▲¥133.489B at the end of 2035. Nominal and discounted payback are not achieved; project IRR is -12.13% and NPV at 8% is ▲¥147.256B.

The largest unresolved drivers are contracted HPC capacity, paid GPU utilization, accelerator-hour price, node delivery cost, fleet OPEX, hydrogen volume/cost, platform pricing, depreciation-heavy investment, working capital, tax, and renewal CAPEX. KPI improvements affect finance only through explicit volume, price, cost, and investment equations.

Node Deployment Plan

Node Deployment Milestone	Cumulative nodes	Objective
By 2030	4	Initial AVP / MaaS validation
By 2032 (scenario)	12	Fleet synchronization
By 2035	36	City-scale operation

KPI and Financial Equation

KPI	Financial impact	Required equation
Utilization	Revenue increase	available vehicle-hours × monetized trips/hour
Fleet idle ratio	OPEX/CAPEX efficiency	idle hours × avoidable fleet cost
Matching rate	Trips increase	requests × matching × dispatch × realization
Dispatch success	Revenue stability	matched trips × successful dispatch
Energy cost/trip	Margin	energy input × unit cost ÷ trips
Node/HPC utilization	CAPEX efficiency	paid GPU-hours ÷ available external GPU-hours
GPU-hour price	Compute revenue	contracted HPC revenue ÷ paid GPU-hours
Trips/day	Revenue scale	fleet × available hours × trips/hour
Load factor	Revenue efficiency	occupied passenger-km ÷ available seat-km
EBITDA margin	Cash generation	EBITDA ÷ revenue
Operating margin	Accounting profit	operating profit ÷ revenue

Screening-Case Scope Check

Completion Check	Status
P&L / cash-flow asset schedules reconciled	Production-case reconciliation required
H2 internal/external cohorts non-overlapping	Production-case cohort split required
Depreciation rows re-summed	YES
IRR recalculated from displayed FCF	YES
NPV at 8% calculated	YES
Nominal project payback calculated	YES
Discounted payback calculated	YES
Node payback separately calculable	Node-level cash flow required
Rendered Word/PDF layout verified	YES
Working capital included	Excluded from screening case
Cash taxes included	Excluded from screening case
Financing structure included	Excluded from screening case
Residual value included	Excluded from screening case
Renewal CAPEX included	Not separately modeled
Node/HPC revenue/OPEX/CAPEX split	Not separately modeled

Investment Decision Logic

- Autonomous-mobility profitability is dominated by utilization, load, price, energy, fleet OPEX, and infrastructure CAPEX - not labor reduction alone.
- Node deployment may convert parking assets into operational infrastructure only after contract, COGS, utilization, permit, and residual-value assumptions are defined.
- AVP-based infrastructure sharing can support private users and MaaS fleets, but revenue cannot be recognized twice across base fares, platform fees, node fees, and settlements.
- Hydrogen-energy integration can reduce OPEX only if metered end-to-end efficiency, production cost, storage, losses, and uptime support the claimed unit cost.
- Vehicle OEMs, mobility-platform providers, operators..., suppliers, and finance entities require contractually explicit authority and accountability; the responsibility structure is proposed, not current fact.
- Distributed-HPC sales improve project returns but do not achieve nominal payback by 2035; contracted capacity, staged investment, lower CAPEX/OPEX, or an alternative financing structure remains necessary.

Financial Driver Targets

Driver	2030 target	2035 target	Financial impact	Evidence status
Utilization	0.75	0.85	primary volume driver	target
Fleet idle ratio	0.25	0.15	OPEX/CAPEX efficiency	target
Energy cost/trip	≤¥500	≤¥350	margin	target
HPC price/GPUh	¥700	¥600	compute revenue	scenario
Matching rate	≥90%	≥95%	demand realization	target
HPC paid utilization	35%	55%	compute revenue	scenario
Trips/day	73	140	revenue scale	scenario
Load factor	0.70	0.80	revenue efficiency	scenario
EBITDA margin	42.4%	35.8%	cash generation	calculated
Operating margin	-36.7%	9.1%	accounting profit	calculated

Investment Assessment

Investment assessment	Target	Displayed case	Status
Nominal project payback	By 2035	Not achieved	BELOW HURDLE
2035 annual FCF	>0	¥40.574B	POSITIVE
IRR	≥12%	-12.13%	BELOW HURDLE
NPV @8%	≥0	▲¥147.256B	BELOW HURDLE
Discounted payback	By 2035	Not achieved	BELOW HURDLE

Distributed-HPC sales improve 2035 annual FCF to ¥40.574B, but cumulative investment is not recovered by 2035; contracted compute revenue and CAPEX/OPEX optimization remain necessary to meet the 8% NPV and 12% IRR hurdles.

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Design Constraints & System Physics - Source Register

Values are classified as Official value, Technical target, Scenario assumption, or Reference only. A source link does not convert a scenario assumption into a fact. Rows with no page-specific or value-specific support are retained as references but must not be used as validated inputs.

Category	Variable	Corrected value / status	Source	Evidence classification / use
Constraint	Insurance requirement reference	Defined only after exact provision identified	MLIT guidance	Reference only
	Operational-time regulation	Define by service/vehicle/operator law	MLIT transport	Reference only
	Safety-margin requirement	Safety-case derived; no generic numeric value	MLIT safety	Reference only
	Traffic regulation	Applicable rules must be mapped to ODD	NPA self-driving	Official framework
	Weather constraint	ODD-specific threshold required	MLIT AD	Reference only
Cost	Charging infrastructure cost	10–50M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	EV cost benchmark	5–10M JPY/unit retained as scenario range	IEA EV Outlook	Scenario assumption
	Hydrogen-station CAPEX	150–300M JPY retained as scenario range	IEA Future of H2	Scenario assumption
	Software cost ratio	10–30% retained as scenario range	OECD finance	Scenario assumption; not source-specific
Demand	Reference-market population	Define after the target geography is selected	Statistics Bureau	Official table; value not repeated here
	Japan population	122.68M at 2026-08-01	STAT_POP_202608	Official value
	Taxi fare	Local fare table required; no national 2,000–2,500 constant	MLIT taxi	Official framework
	Trips per person	2.3/day retained only as survey assumption	MLIT transport survey	Scenario assumption
	Target-region vehicle stock	Use latest Chubu table	Chubu vehicle stats	Official table; validate date
Energy	Battery storage cost trend	Declining; exact year/system boundary required	NREL battery	Reference only
	Electricity intraday price	13.0 JPY/kWh retained only with period definition	METI_ELEC_p14	Scenario/period-specific
	Electricity retail price	30.2 JPY/kWh	METI_ELEC_p14	Official cited benchmark
	Electricity spot price	12.24 JPY/kWh retained only with period definition	METI_ELEC_p14	Period-specific
	Hydrogen LHV	33.3 kWh/kg	DOE_H2_LHV	Physical constant
	Hydrogen pathway roundtrip efficiency	Lower than direct electricity; system boundary required	NREL H2 pathway	Reference only
	Hydrogen production cost	150–200 JPY/kg retained as scenario	IEA Future of H2	Scenario assumption
	Hydrogen sales price	300–400 JPY/kg retained as scenario	NEDO H2 library	Scenario assumption
	Hydrogen specific energy	120 MJ/kg	DOE_H2_LHV	Physical constant
	Hydrogen storage cost	DOE 2020 target: $333/kg-H2 stored	DOE_2020_STORAGE	Technical target; not achieved current value
	Hydrogen storage gravimetric	DOE 2020 target: 4.5 wt%	DOE_2020_STORAGE	Technical target; not achieved current value
	PEMFC direct-H2 efficiency	~60% LHV	DOE_PEMFC_60	Technology comparison; not 22%
	LNG energy density	13.9 kWh/kg retained	IEA Future of H2	Reference; page/definition required
	LNG volumetric energy	20.8 MJ/L retained	IEA Future of H2	Reference; page/definition required
	Liquid-H2 volumetric energy	~8 MJ/L retained	DOE H2 storage	Reference
	Generation cost benchmark	LCOE reference only	IEA generation cost	Reference only
	PV high-output benchmark	>1,100kWh/kW/yr	NEDO_PV_p14	Benchmark
	PV output benchmark	1,000–1,100kWh/kW/yr	NEDO_PV_p14	Benchmark
	SOFC efficiency	~60% scenario benchmark	DOE fuel cells	Technology-dependent
Financial	Discount rate	8–12% retained as screening range	OECD finance	Scenario assumption
	Infrastructure return	Project-specific	JBIC	Reference only
	Investment hurdle	Project-specific	OECD finance	Reference only
	Project IRR benchmark	10–15% retained as hurdle range	JBIC	Scenario assumption
Infra	H100 cloud price benchmark	Public cloud ≈$11/GPUh; model ¥400–900/h	Google A3	Benchmark; model is scenario
	GPU/system power	H100 up to 700W; model 1.2kW/GPU + PUE 1.15	NVIDIA H100	Official GPU limit + scenario
	Hydrogen infrastructure	Definition/reference only	IEA Future of H2	Reference only
Logistics	Standard freight fare	Distance/time and vehicle-class tables; no national 52 JPY/kg constant	MLIT_STANDARD_FARE	Official framework
Market	Global EV sales share	15% retained only with year/definition	IEA EV Outlook	Period-specific
	Heavy-truck new sales (Japan)	73k retained only after exact JAMA table validation	JAMA statistics	Validate year/category
	Japan EV sales share	1% retained only with year/definition	IEA EV Outlook	Period-specific
	Total new vehicle sales (Japan)	4.421M retained only after exact JAMA table validation	JAMA statistics	Validate year/category
Policy	Japan 2035 electrified-vehicle target	100% new passenger-car sales; scope definition required	Public policy context	Policy definition required
	Electrified-vehicle definition	BEV/PHEV/HEV/FCEV	METI energy policy	Policy definition
	Japan powertrain distribution	Use a dated JAMA powertrain table	JAMA statistics	Reference only
Regulation	Accident response obligation	Required	NPA_L4	Official
	Administrative cancellation	Applicable	NPA_L4	Official
	Autonomous-driving subsidy availability	Program-specific	MLIT subsidy	Verify FY/eligible costs
	Current safety-policy baseline	Applicable	MLIT safety	Official framework
	Government safety policy	Defined	MLIT AD	Official framework
	Human-operated-vigilance requirement	Depends on operation type	NPA_L4	Official framework
	Infrastructure regulation	Defined by installed equipment/site	MLIT AD	Map exact law
	Insurance requirement	Required; exact coverage required	MLIT AD	Map exact provision
	International standards relevance	Relevant, not automatically legally binding	MLIT safety	Reference
	Legal-liability framework	Defined across product/operator/road law	MLIT safety	Map exact provisions
	Level 4 summary availability	Available	NPA_L4	Official
	Local-government consultation	Required where specified	NPA_L4	Official
	Monitoring requirement	Required where specified	NPA_L4	Official
	Non-specific operation treatment	Defined	NPA interpretation	Official interpretation
	ODD requirement	Required	NPA_L4	Official
	Operation plan submission	Required	NPA portal	Official
	Passenger transport rule	Applicable by service model	e-Gov Passenger Rule	Official law
	Permit cancellation	Applicable	NPA_L4	Official
	Police reporting	Required where specified	NPA_L4	Official
	Public-road operation framework	Defined	MLIT AD	Official
	Remote monitoring	Allowed under defined conditions	NPA_L4	Official
	Road Traffic Act	Applicable	e-Gov Road Traffic Act	Official law
	Road Transport Vehicle Act	Applicable	e-Gov Vehicle Act	Official law
	Road-use regulation	Defined	MLIT safety	Official
	Running-environment condition	Required	MLIT AD	Official
	Specific autonomous-operation permit	Required	NPA portal	Official
	Subsidy eligible cost	Program-specific	MLIT subsidy	Verify current notice
	Suspension authority	Applicable	NPA_L4	Official
	Transport safety law reference	Defined	MLIT safety	Official
	Vehicle Act enforcement order	Applicable	e-Gov order	Official law

Note on Information Sources and Intellectual Property

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